

Out-of-Sample Workload Feasibility in CSLAP Layouts: an Empirical Study and a Budgeted-Uncertainty Robust Counterpart

CSLAP research pipeline

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Abstract

We audit the workload-feasibility semantics of the CSLAP evaluation pipeline and re-analyze every stored layout of the project’s instance families (iscf480, iscf10kt, iscfco, planted-affinity, BERNER). The as-run out-of-sample check compares realized test workload against a *train-volume-calibrated* ceiling; its bindingness B collapses from ≈ 1.0 down to 0.11–0.13 across expanding temporal cuts, so late-cut “**wl_broken=0**” verdicts are volume slack, not robustness. Under a volume-normalized ceiling the picture reverses: every temporal fold carries 1–3 of 8 stations above the fair-share ceiling at $k=1$ (1–4 over the full closure grid), with peak load ratios up to 2.52 at $k=1$ (2.63 over the full grid; 3.45 in the sparse-train variants), while the cross-validation folds remain balanced to the exact 10%-slack edge. A retrospective Bertsimas–Sim calibration shows the realized overloads sit in the budget-uncertainty sweet spot: median Γ^{cover} of 8/60 and 19/541 assigned products, deviations head-concentrated (median Gini 0.63/0.82), and $\Gamma^{\text{bind}} = 1$ at most stations. We derive the exact linear robust counterpart of the workload constraint and implement it (HiGHS plus a deterministic local search, paired over the temporal cuts). Full-magnitude protection provably does not fit the 10%-slack capacity contract (infeasibility certificates at $\Gamma \geq 4$); ex-ante half-magnitude protection at $\Gamma = 1$ cuts the out-of-sample peak ratio from 1.77 to 1.59 and the excess workload from 17.4% to 13.0% for a 2.3% visit price, while uniform ceiling-tightening at the same price buys no peak reduction at all — targeted budgeted protection dominates its nominal competitor.

1 Introduction and context

The Correlated Storage Location Assignment Problem (CSLAP) assigns products $p \in P$ to picking stations $s \in S$ (binary x_{ps}) so as to minimize the total number of station visits over a set of orders, subject to a slot-capacity constraint $\sum_p x_{ps} \leq \zeta_s$ and a workload constraint $\sum_p x_{ps} L_p / V_s \leq T_s$ per station. The project’s preceding research arc took the covering-enlargement hypothesis H1 — replace each order by its pattern-closure and optimize placement on the enlarged instance to buy out-of-sample robustness *for the visits objective* — end-to-end through an exact CPLEX re-validation, and falsified it: the price of robustness is real and the out-of-sample gain is negative, with confidence intervals excluding zero.¹

The same arc, however, surfaced one instability in the data that is *not* an artifact of the covering mechanism: *demand drift*. The drift-characteristics and order-data diagnostics² showed that per-product line counts move substantially between a training window and a later operating window,

¹Internal reports docs/background/7_scientific_report_iscf_cplex.md through docs/background/11_bernier_decomposition.md in this repository.

²docs/background/8_drift_characteristics.md and docs/background/9_order_data_diagnostics.md.

while the placement pipeline calibrates its workload ceiling once, on the training window, and never revisits it. This study therefore opens a different channel from H1: not the visits objective, but **out-of-sample workload feasibility**. The research question is:

Does a CSLAP layout that satisfies the (hard, train-enforced) workload constraint remain workload-feasible on later demand, and if not, what deviation magnitudes and protection budgets would a budgeted-uncertainty robust workload constraint have needed?

Contributions:

- **Pipeline audit** (Section 2): precise semantics of the as-run feasibility check, and a *bindingness* index B showing the check degenerates into a volume-calibration test on expanding temporal cuts.
- **Volume-normalized reanalysis** (Section 3): a counterfactual fair-share ceiling that removes the volume artifact, applied to all stored layouts across six ISCF-derived families plus the industrial BERNER cell; every recomputed as-run count reproduces the stored `wl_broken` exactly (276 result cells checked across families), and the train-side constraint held in every one of the $> 1,000$ station-checks — so all reported violations are genuinely out-of-sample.
- **Retrospective Bertsimas–Sim calibration** (Section 3.8): three deviation-magnitude models (M1 proportional, M2 square-root, M3 train-internal empirical), fitted coverage constants, and the diagnostics $\Gamma^{\text{cover}} / \Gamma^{\text{bind}}$ quantifying how large a budget the observed drift asks for and how small a budget already forces retraining.
- **Robust counterpart** (Section 4): the exact linear Bertsimas–Sim counterpart [1] of the CSLAP workload constraint, in the row-wise framework of [4], with the Soyster box model [3] as its $\Gamma \rightarrow \zeta_s$ limit.
- **Implementation and paired temporal experiment** (Section 5): the counterpart implemented on HiGHS with a deterministic local-search incumbent pipeline and evaluated, paired arm-by-arm, over the five temporal cuts — yielding infeasibility certificates for full-magnitude protection under the 10%-slack contract, a price-severity frontier for partial insurance, and the head-to-head result that budgeted protection dominates equal-price uniform ceiling-tightening on out-of-sample peak severity.

2 The current evaluation pipeline

Placement (train side). Layouts are produced by the Hexaly placement model, which enforces the workload constraint *hard* on training data:³ for every station s , $\sum_p (L_p/V_s) x_{ps} \leq T_s$, where L_p is the product’s training-window line count (`REAL_LINES`, distinct orders containing p). Our recomputation confirms the constraint held on the training data for every stored layout of every family (`viol_train=0` in all 800 `iscf10kt` station-rows, all 160 gamma-family checks, and all 1,088 secondary station-rows).

³`milp_synthetic.py`, lines 70–77; the line counts are passed by the harness at lines 427 and 441 of `run_robustness_iscf.py`.

Ceiling calibration. The time capacity is set once per fold from training volume by the 10%-slack rule (`run_robustness_iscf.py`, line 211; same formula in `iscf_adapter.py`, line 194):

$$T_s = \left\lceil \sigma \cdot \frac{\sum_p L_p^{\text{tr}}}{V_s |S|} \right\rceil, \quad \sigma = 1.10, \quad (1)$$

with V_s the station speed ($V_s = 1$, uniform, on all ISCF families; $\sigma = 1.20$ and heterogeneous per-station ceilings on BERNER).

Evaluation (test side). The test instance re-uses the *train* stations file verbatim.⁴ The evaluator (`evaluate_layout_robust.py`, lines 129–142, driven at `run_robustness_iscf.py`, lines 490–504) computes the test demand ℓ_p^{te} of each product as its raw test line count, i.e. `groupby('PRODUCT').size()` over the test orders; the realized workload $W_s^{\text{te}} = \sum_{p \in \Phi_s} \ell_p^{\text{te}} / V_s$ over the assigned set Φ_s of the stored layout (a flat JSON dict `{PROD_<id>: GARE<n>}`); and the count `wl_broken = |\{s : W_s^{\text{te}} > T_s\}|` (strict inequality) and `cap_broken` analogously against the slot count. Temporal cuts are 5 expanding-train splits at train fractions 0.5, 0.6, ..., 0.9 by `ORDER_ID` rank (train-early/test-late); the cuts share train mass and are a dispersion proxy, not independent replicates.

The volume artifact. The as-run check thus compares *test-window* workload against a *train-volume-calibrated* ceiling. Define the **bindingness** of the check as

$$B = \frac{\sum_p \ell_p^{\text{te}}}{\sum_s V_s T_s}, \quad (2)$$

the ratio of total test volume to total ceiling capacity. With uniform speeds and a flat T_s , a layout balanced exactly to the slack edge (peak load = $\sigma \times$ fair share) crosses T_s if and only if $B > 1/\sigma \approx 0.909$; below that threshold the check cannot fire on any near-balanced layout, regardless of how unbalanced the demand mix has become. On the expanding temporal design the test window shrinks as the cut moves late, so B decays mechanically (from 1.005 to 0.125 on `iscf480`) and the as-run verdict degenerates into a measurement of window length. Conversely, when the test window carries *more* volume than the train window (sparse-train and affinity families, $B > 1$), even a perfectly balanced layout must violate, and the as-run count saturates at 8/8 for reasons that have nothing to do with layout quality.

3 Empirical study

3.1 Datasets and stored layouts

Table 1 lists the families. The affinity scale cell quotes the rep-r0 arms (unique test orders 3,645–3,986; over all 18 arms the range is 3,645–4,000). The affinity replicate r1 is degenerate as-run (all 12 layout JSONs are empty, stored rows all-zero), leaving 24 informative arms; BERNER persisted no layouts, so its audit (Section 3.7) bounds bindingness through assignment assumptions rather than recomputing realized loads.

⁴`run_robustness_iscf.py`, lines 229–233.

Table 1: Instance families audited. All ISCF-derived families have 8 uniform stations (GARE4–GARE11, $V_s = 1$, $\sigma = 1.10$); BERNER has 24 heterogeneous stations ($\sigma = 1.20$). k is the pattern-closure order of the stored layout ($k = 1 = \text{nominal}$).

Family	$ P $	ζ_s	Split design	k grid	Scale (fold 0)
iscf480 CV	480	60	5 random folds	1, 2	246.7k / 61.7k orders
iscf480 temporal	480	60	5 expanding cuts	1, 2	154,207 / 154,208 orders
iscf10kt temporal	4,328	541	5 expanding cuts	1, 2, 3, 4, 6	5,000 / 5,000 orders
iscf10kt t500/t1000/t2000	4,328	541	train capped	1, 2, 3, 4, 6	500–2,000 / 1,000–5,000
iscfco mm/ms	1,600	200	5 expanding cuts	1, 10	7,500 / 7,500 orders
planted affinity	800	100	18 arms (2 modes, 3 α , 3 reps)	1, 10	1,847 / 3,645–3,986
BERNER	2,000	9–207	1 temporal cut (10/13 wk)	1–20	778,788 / 220,893 lines

3.2 Metrics

Let $L^{\text{tr}} = \sum_p L_p^{\text{tr}}$ and $L^{\text{te}} = \sum_p \ell_p^{\text{te}}$ be total train and test lines, and $\lambda = L^{\text{tr}}/L^{\text{te}}$ the volume ratio. Consistent with the notation of Section 4:

- **As-run check:** $W_s^{\text{te}} > T_s$; count `viol_asrun = wl_broken`.
- **Normalized check:** the volume-normalized load is $W_s^n = \lambda W_s^{\text{te}}$ (equivalently, the raw load is compared against the counterfactual test-volume ceiling $T_s^n = \lceil \sigma L^{\text{te}} / (V_s |S|) \rceil$; the two violation sets coincide on every audited fold). We report `viol_norm` = $|\{s : W_s^n > T_s\}|$ and the peak ratio $\text{ratio}^n = \max_s W_s^{\text{te}}/T_s^n$.
- **Imbalance:** $\max_s W_s^{\text{te}} / (L^{\text{te}} / (V_s |S|))$, the peak-to-fair-share ratio; values $\leq \sigma = 1.10$ are within the slack rule.
- **Bindingness** B as in Equation (2); volume-binding threshold $1/\sigma \approx 0.909$.
- **Excess fraction** $X = \sum_s (W_s^n - T_s) / \sum_s W_s^n$, the share of (normalized) test workload sitting above the ceiling.

Sanity. The recomputation reproduces the stored `wl_broken` in *all* cells: 20 (split, fold, k) cells on iscf480, 100 (variant, fold, k) rows on iscf10kt, 136 secondary cells (20 iscfco base + 36 affinity + 80 iscfco demand-shift), and the 20 cells of the Γ -calibration re-run; recomputed peak workloads match the stored `max_workload` to $< 10^{-6}$ (e.g., iscf480 temporal fold 0, $k = 1$: 86,019 vs $T_s = 45,365$). `cap_broken=0` everywhere.

3.3 Primary instance: iscf480 (CV vs. temporal)

The CV anchor is a volume artifact. Table 2 shows that the headline “`wl_broken=0` on all CV folds” says nothing about balance: with $B \approx 0.227\text{--}0.228$ against a threshold of 0.909 (Figure 2), the train-calibrated ceiling is roughly $4\times$ too loose for the CV test volume. The normalized check does surface violations in 4 of 5 folds at $k=1$ — because Hexaly balances the training workload to exactly the 10%-slack edge (imbalance 1.09–1.12 across all (fold, k)), any drift tips the peak station over — but they are *marginal*: peak ratio at most 1.016, excess fraction at most 0.40%. The honest CV statement is “balanced to the slack edge, no material overload.”

Temporal drift is severe. Under the temporal design (Table 3), every cut has 2–3 of 8 stations above the normalized ceiling at $k=1$ (2–4 over both k), with peak load ratios 1.71–2.52 and $X = 11.0\text{--}30.7\%$ of total test workload above the ceiling; imbalance reaches 2.77 (cut r0f3, $k=1$).

Table 2: iscf480, 5-fold random CV. The as-run anchor `wl_broken=0` everywhere is reproduced, but $B \approx 0.23$: the test window carries only $\sim 23\%$ of the volume the ceiling assumes. Normalized violations exist but are hair-thin (peak ratio ≤ 1.016 , $X \leq 0.40\%$). Source: `results/iscf480/per_fold.csv`.

Fold	B	as-run $k=1$	as-run $k=2$	norm $k=1$	norm $k=2$	ratio ⁿ $k=1$	X (%) $k=1$
r0f0	0.228	0	0	2	2	1.016	0.24
r0f1	0.227	0	0	0	0	0.999	0.00
r0f2	0.227	0	0	2	1	1.005	0.13
r0f3	0.227	0	0	3	1	1.012	0.40
r0f4	0.227	0	0	1	0	1.000	0.00

Table 3: iscf480, 5 expanding temporal cuts. As-run counts merely track the decaying bindingness B ; the normalized check reveals 2–4 of 8 stations overloaded in *every* cut, with peak ratios up to 2.52. Violated-station counts of 8 stations. Source: `results/iscf480/per_fold.csv`.

Cut	B	as-run ($k=1$ / $k=2$)		norm ($k=1$ / $k=2$)		ratio ⁿ ($k=1$ / $k=2$)	
r0f0	1.005	4	2	3	2	1.71	1.54
r0f1	0.693	2	1	3	4	1.91	1.71
r0f2	0.457	1	0	3	3	2.25	1.98
r0f3	0.281	0	0	2	3	2.52	2.19
r0f4	0.125	0	0	3	3	2.38	1.88

Meanwhile the as-run count decays 4, 2, 1, 0, 0 in lockstep with B ($1.005 \rightarrow 0.125$; Figure 1): the late cuts look “clean” as-run exactly where the true overloads are worst. Only cut r0f0 is volume-binding ($B = 1.005$), and there the as-run and normalized checks agree (4 vs 3 stations). The identity of the peak station changes across cuts (GARE10, GARE4, GARE11, GARE11, GARE6 at $k=1$; Figure 3; top-3 detail in `results/iscf480/top3_overloaded_temporal.csv`); every station holds exactly 60 products in every layout (slot capacity fully binding), so the overloads are pure demand-mix drift, not slot-count effects. The $k=2$ closure layouts are consistently *less extreme* at the peak (mean peak ratio 1.86 vs 2.15; lower in all 5 cuts) but remain far infeasible, and do not reduce the violation *count* (cuts r0f1, r0f3).

3.4 Full-catalogue instance: iscf10kt temporal and sparse-train variants

Table 4: iscf10kt temporal ($k \in \{1, 2\}$; full k grid in Table 14). Same two-regime pattern as iscf480. Counts of 8 stations. Source: `results/iscf10kt/per_fold.csv`.

Cut	B	as-run ($k=1$ / $k=2$)		norm ($k=1$ / $k=2$)		ratio ⁿ $k=1$	X (%) $k=1$
0	0.786	2	2	2	2	1.50	10.6
1	0.550	1	0	2	2	1.69	15.6
2	0.367	0	0	1	2	2.15	15.8
3	0.224	0	0	3	1	1.93	24.6
4	0.114	0	0	3	2	2.35	25.4

On the full 4,328-SKU catalogue over a coverage-guaranteed 10k-order subsample (Table 4), the stored as-run anchor ($k=1$: 2, 1, 0, 0, 0) is reproduced exactly (fold 0 peak 2,464 vs $T_s = 1,904$,

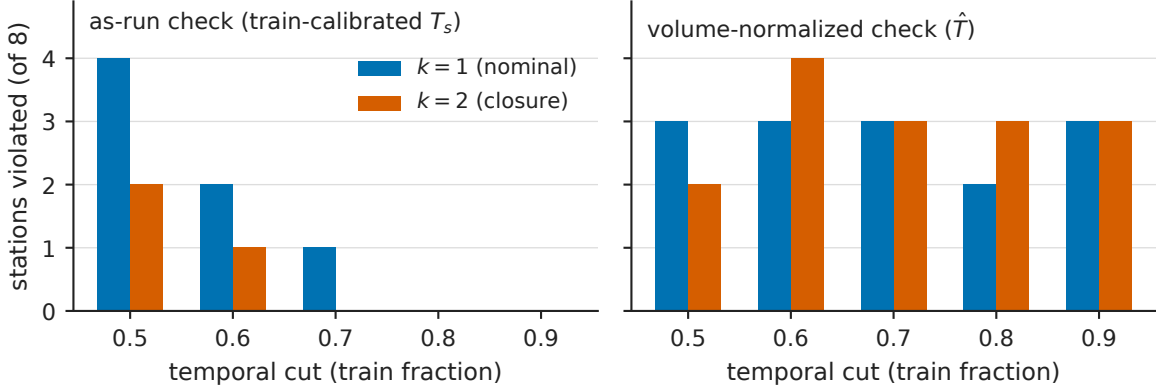


Figure 1: iscf480 temporal cuts: as-run vs volume-normalized violated-station counts (of 8). At $k=1$ the as-run count decays 4, 2, 1, 0, 0 while the normalized count stays at 3, 3, 3, 2, 3; at $k=2$: as-run 2, 1, 0, 0, 0 vs normalized 2, 4, 3, 3, 3. The as-run decay tracks test-window volume, not balance. Source: `results/iscf480/per_fold.csv`.

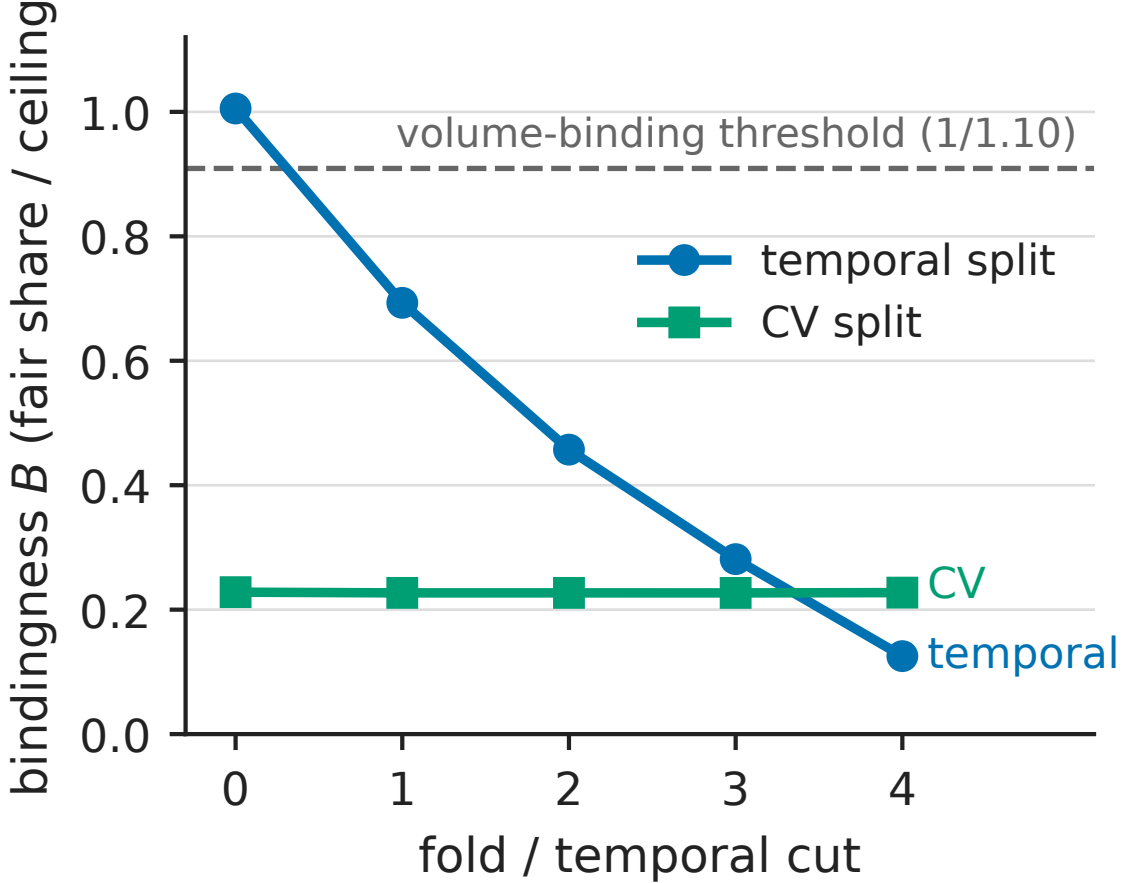


Figure 2: Bindingness B of the as-run check across iscf480 folds. Temporal B falls $1.005 \rightarrow 0.125$ as the expanding cut shrinks the test window, while CV B is flat at ≈ 0.227 ; both are judged against the volume-binding threshold $1/1.10 \approx 0.909$. Only temporal cut 0 is volume-binding; everywhere else the as-run check is structurally unable to fire on a near-balanced layout.

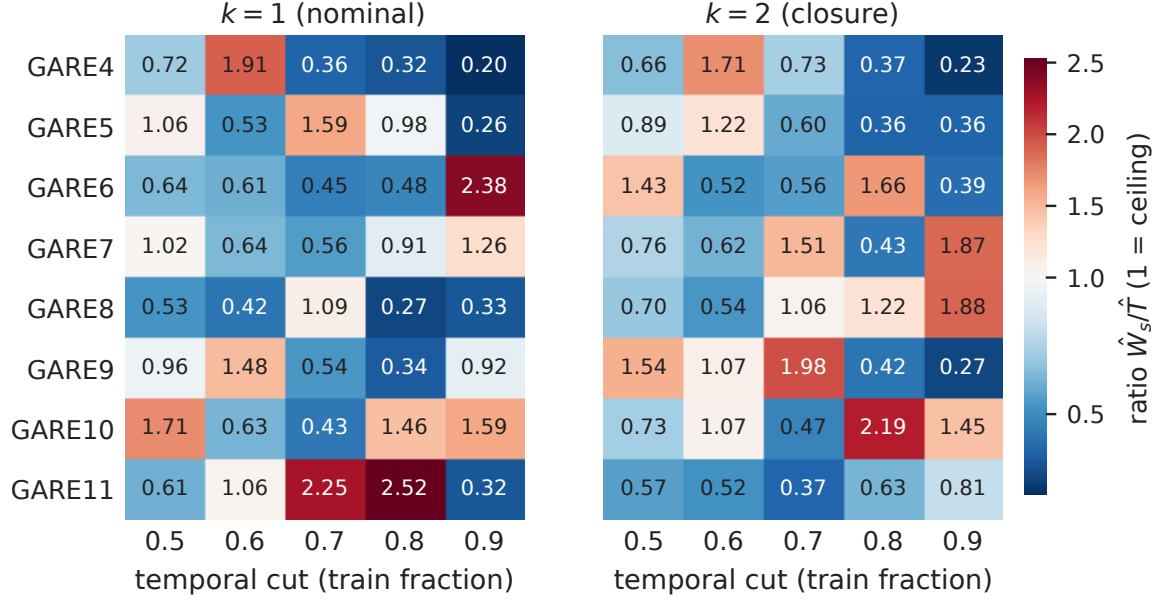


Figure 3: Station-level normalized load ratios $W_s^{\text{te}}/T_s^{\text{n}}$ for iscf480 temporal, all 8 stations $\times$ 5 cuts, $k=1$ (top) and $k=2$ (bottom); diverging scale centered at 1.0 (ratio > 1 = overloaded under fair-share volume), all 80 cells annotated. Data range 0.20–2.52 at $k=1$; $k=2$ peaks at 2.19. The overloaded station moves across cuts. Source: `results/iscf480/heatmap_temporal_k1.csv` and `..._k2.csv`.

as-run ratio 1.29), and again B falls $0.79 \rightarrow 0.11$, all below the 0.909 threshold — the fold-4 ceiling is $\approx 8.8\times$ the fair-share test volume. Normalized, *all 100* (variant, fold, k) stored layouts of the iscf10kt families violate on 1–3 of 8 stations; imbalance ranges 1.43–3.80 (always above the 1.10 slack rule) and $X = 6.6$ –37.5%.

Table 5: Sparse-train effect (iscf10kt, means over 5 folds $\times k \in \{1, 2, 3, 4, 6\}$ unless marked $k=1$). The dramatic as-run violation counts at small train samples are a ceiling artifact ($B \gg 1$: with more test volume than total ceiling capacity even a perfectly balanced layout must violate); the volume-normalized violation count is flat, and genuine degradation shows only in the $k=1$ imbalance. Source: `results/iscf10kt/sparse_train_effect.csv`.

Variant	train orders	as-run viol.	B	norm viol.	ratio ⁿ	X (%)	imbalance ($k=1$)
t500	500	7.56	5.06	2.24	2.06	17.7	2.55
t1000	1,000	6.48	2.53	2.16	1.94	17.0	2.34
t2000	2,000	4.52	1.27	2.44	1.92	18.9	2.08
temporal (full)	7,000	0.40	0.41	2.16	1.86	17.1	2.12

Table 5 makes the artifact direction explicit in reverse: at t500 the as-run check breaks a mean 7.56 of 8 stations ($k=1$ by fold: 8, 8, 8, 8, 4), but $B = 2.07$ –8.18 there, so those counts measure train-volume calibration, not placement quality. Normalized, the violation count is essentially flat across train sizes (2.24 / 2.16 / 2.44 / 2.16), and the genuine sparse-data cost appears only as $k=1$ imbalance growing from 2.12 (full) to 2.55 (t500). The single worst layout overall is t500 fold 4 $k=1$: peak ratioⁿ 3.45, imbalance 3.80 — one station carries 47.5% of test workload.

3.5 Effect of closure order k on balance

Table 6: k -effect on the iscf10kt temporal variant (means over 5 cuts). Closure enlargement smooths the peak slightly but never approaches feasibility, and the violation count does not improve monotonically. Source: `results/iscf10kt/k_effect.csv`.

	$k=1$	$k=2$	$k=3$	$k=4$	$k=6$
mean norm violations (of 8)	2.2	1.8	1.8	2.4	2.6
mean imbalance	2.12	2.13	2.11	2.03	1.85

Across all 20 (variant, fold) pairs of the iscf10kt families, *some* $k > 1$ beats $k=1$ on imbalance in 17/20 pairs (mean best- k gain 0.43 imbalance points over the 20 pairs), but the best k is unstable (2, 3, 4 and 6 all occur) and the normalized violation count never reaches 0 (Table 6, Figure 4). On iscf480 temporal the same holds: $k=2$ lowers the peak ratio in all 5 cuts (mean 1.86 vs 2.15) without reducing the violation count. Closures *spread* the overload rather than remove it — consistent with the H1 falsification, enlargement is not a workload-feasibility mechanism.

3.6 Secondary families and planted shifts

iscfco base (co-occurrence-rich, 1,600 SKUs). The as-run check reports `wl_broken=0` in 19 of 20 arms (the exception, ms fold 0 $k=10$, has 1 station at as-run ratio 1.001), yet 1–3 of 8 stations breach the normalized ceiling in *every* arm: imbalance 1.12–1.20 (always above 1.10), peak ratioⁿ 1.01–1.09, $X = 0.2$ –2.0%. Bindingness collapses $0.858 \rightarrow 0.099$ across cuts, so the late-cut zeros are again volume slack. This is the mild edge of the same phenomenon: material but small overloads on an instance with denser co-occurrence structure.

Planted demand shift (iscfco, the one mode that moves per-product volumes). Table 7 summarizes the only shift family whose stations file carries a volume-recalibrated ceiling ($B = 0.907$ – $0.909 \approx 1/1.10$, i.e., the as-run demand check is volume-binding by construction, and as-run $\approx$ normalized there). The nominal $k=1$ layout violates on 1–4 of 8 stations at *every* shift intensity ρ , including $\rho = 0$; the violation count is flat in ρ but the severity scales: mean peak ratio $1.07 \rightarrow 1.33$ and X $1.8\% \rightarrow 6.9\%$ from $\rho = 0$ to $\rho = 2$. The $k=10$ closure layouts show the same gradient (X $1.0\% \rightarrow 5.6\%$) — covering enlargement does not repair demand-drift infeasibility.

Table 7: iscfco demand shift, means over 5 cuts (normalized metrics; $k=1$ layouts are identical across the mm/ms variants and deduplicated; $k=10$ pooled over mm and ms). Source: `results/secondary/per_fold.csv`.

ρ	norm viol. $k=1$ (of 8)	ratio ⁿ $k=1$	X (%) $k=1$	X (%) $k=10$
0.0	2.6	1.07	1.8	1.0
0.5	2.2	1.10	2.5	2.3
1.0	2.2	1.19	3.7	2.7
2.0	2.6	1.33	6.9	5.6

Planted affinity drift. The stored as-run verdicts (7–8 of 8 stations broken in all 24 informative arms, as-run ratio up to 5.25) are almost entirely volume artifact: test mass is 1.79 – $4.48\times$ the

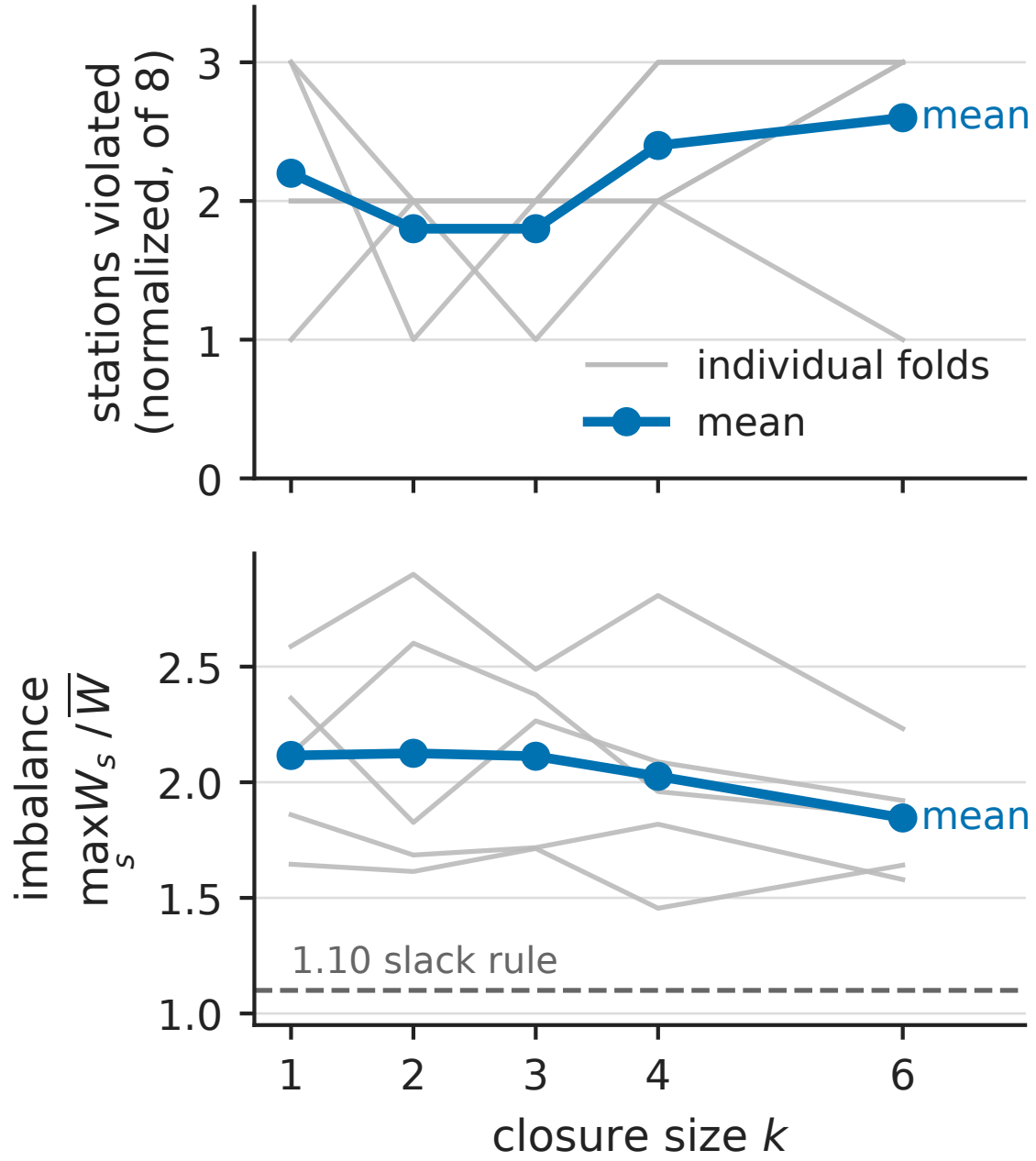


Figure 4: k -effect panels, iscf10kt temporal variant, $k \in \{1, 2, 3, 4, 6\}$: mean normalized violations stay between 1.8 and 2.6 of 8 stations, and mean imbalance eases only from 2.12 to 1.85 — never near the 1.10 slack rule. Source: `results/iscf10kt/k_effect.csv`.

train-calibrated ceiling ($B > 1$). Normalized, the genuine violations are 1–4 of 8 (mean 2.29), imbalance 1.14–1.37, peak ratioⁿ up to 1.24 — and they grow with planted drift: the reassign arms at $\alpha = 0.5 / 1.0$ average imbalance 1.318 / 1.285 and $X = 4.8\% / 3.6\%$ vs ≈ 1.21 – 1.22 and $\approx 2\%$ at $\alpha = 0$. Full per-arm rows in Table 17.

As-run aggregate over all stored shift evaluations. Pooling the 1,660 stored rows of the 18 `per_fold_shift_*.csv` files (Table 8), `wl_broken` > 0 in 969 rows (58.4%) and `cap_broken`=0 in all 1,660. The split by family confirms the diagnosis: the sparse-train families are broken in 100% of rows (ceiling artifact, $B \gg 1$), the full-train `iscf10kt` families in ~ 25 – 30% (mean 0.40 of 8), the structured/unstructured `iscfco` shifts almost never (mm 0/80, ms 8/80 with max 1, i.e. 4/40 in each of the structured and unstructured modes) — both shift modes preserve per-product volumes — while the demand mode breaks 80/80 rows. As-run counts are essentially ρ -insensitive; only volume calibration and the demand mode matter.

Table 8: As-run aggregate of all stored shift evaluations (per results directory). “% rows broken” = share of rows with `wl_broken` > 0. Source: `results/secondary/shift_asrun_aggregate.csv`.

Results dir	rows	% rows broken	mean <code>wl_broken</code>	max
<code>iscf10kt_results_mm</code>	260	30.0	0.40	2
<code>iscf10kt_results_ms</code>	260	25.0	0.40	2
<code>iscf10kt_results_shift</code>	225	28.0	0.40	2
<code>iscf10kt_results_t500</code>	225	100.0	7.56	8
<code>iscf10kt_results_t1000</code>	225	100.0	6.48	8
<code>iscf10kt_results_t2000</code>	225	100.0	4.52	8
<code>iscfco_results_mm</code>	120	33.3	0.82	4
<code>iscfco_results_ms</code>	120	40.0	0.82	4

3.7 Industrial cell: BERNER audit

BERNER is the one heterogeneous family: 24 stations with per-station $T_s \in \{1, \dots, 158\}$ (calibrated at 120% of the pinned-assignment train workload, verified exactly for all 24 stations), speeds 332.6–11,885.7, and a single temporal cut (train quantile 0.7692, 10 of 13 weeks; 778,788 raw train lines, 220,893 raw test lines). No layouts were persisted, so realized per-station test loads cannot be recomputed; the audit instead bounds the check’s informativeness four ways, all agreeing:

- **Aggregate bindingness:** $B_{\text{agg}}^{\text{te}} = 220,893/898,472 = 0.246$, far below the $1/1.20 = 0.833$ threshold (the 3-week test window carries 30.1% of the distinct-line `REAL_LINES` train volume of 734,229 — the basis that calibrates T_s — and 28.4% of the 778,788 raw train lines). The stored test-demand `wl_broken`=0 for the deployable arms A/B (all 16 decompose rows, every k) is therefore *volume-trivial*.
- **Assignment bounds:** under the pinned identity assignment the worst station reaches only ratio 0.286 on test demand (train-frequency-proportional split: 0.251) — no train-like layout could plausibly have fired the as-run test check.
- **Train side, the contract is binding:** $B_{\text{agg}}^{\text{tr}} = 734,229/898,472 = 0.817 \approx 1/1.20$ (the train side of the aggregate bindingness uses the pinned-assignment `REAL_LINES` workload, the quantity T_s was calibrated on; the 778,788 raw train lines would give 0.867), pinned train

ratios reach 0.8333 at every large station, and the hard-constrained clean-comparison arms sit at `real_wl_max_ratio` 0.998–1.000 — placement is *at* the ceiling.

- **Shape-bindingness:** a fluid equal-workload split of test demand ($W^* = 5.674$ per station) would overload station 12 ($T_s = 1$) by $5.67\times$ and station 24 ($T_s = 2$) by $2.84\times$: equal balancing is incompatible with the heterogeneous per-station contract, so the constraint binds in shape as well as volume.

Table 9: BERNER feasibility semantics by arm group. “Test” = TEST raw-line demand vs real train stations (volume-trivial, $B_{\text{agg}}^{\text{te}} = 0.246$); “train” = TRAIN-demand (`REAL_LINES`) deployability recheck vs the real per-station contract. Ranges over the k grid of each source CSV. Source: `results/berner/berner_feasibility_audit.csv`.

Arm group	Placement contract	Check	k grid	<code>wl_broken</code> /24	max ratio
A, B (decompose)	real, hard	test	1–10	0	—
C, D (decompose, recal)	loosened flat	test	2–10	0	—
C', D' (cprime)	loose recal + $\bar{L}$ signal	train	2–10	6–9	6.326
B (clean, real regime)	real, hard	train	1–20	0	1.000
C'', D'' (fair-inflated)	per-station fair T_s^{fair}	train	2–20	4–8	1.793
oldD (loose flat cap)	flat eq.-10 recal	train	2–20	7–10	2.810

Table 9 separates the two check semantics. The loose-recalibrated C'/D' arms are confirmed non-deployable: 6–9 of 24 stations breach the *real* train-demand contract, with maximum ratio 5.488 for C' at $k=2$ (matching the “up to $5.5\times$ ” figure quoted in the decomposition report) and an overall maximum of 6.326 (D', $k=4$). The hard-constrained A/B arms break 0 stations at every k . Note the caveat: BERNER order CSVs contain duplicate (ORDER, PRODUCT) rows (18,367 of 220,893 test rows); the as-run semantics counts raw lines, and this audit follows it. Switching to distinct-order test demand moves the headline quantities little — the worst-station pinned test ratio falls from 0.286 to 0.260 (−9.2%) and the aggregate test volume by 8.3% — so no conclusion changes, though individual small stations move more (the degenerate $T_s = 1$ station 12 falls from 0.274 to 0.131).

3.8 Retrospective Γ calibration

We instantiate the estimators of Section 4.5 on the two temporal families (iscf480, iscf10kt; $k=1$ primary, $k=2$ secondary), with deviations $\delta_p = L_p^n - L_p^0$ of volume-normalized test line counts against the train nominal (λ -normalization removes volume drift, so all findings isolate demand-*mix* drift). The recomputation reproduces every as-run anchor and the normalized violation sets exactly (no rounding mismatches; 160 train-side checks all feasible).

Magnitudes (Table 10). The fitted constants are large: median $c_1(0.9) = 1.18$ (iscf480) and 1.63 (iscf10kt), i.e., covering 90% of train-line-weighted realized deviations requires a deviation allowance of roughly 118–163% of a product’s nominal line count, and c_1 grows with the cut horizon ($0.76 \rightarrow 1.48$ and $1.31 \rightarrow 3.20$) — a single calibration is horizon-dependent. The train-internal M3, the only genuinely planner-available model, covers just 1.5–5.4% of the weighted realized deviations on iscf480 (38–80% on iscf10kt): within-train block variability drastically underestimates train→test drift. These constants are fitted *retrospectively* to realized deviations; they quantify what would have been needed, not what a planner could estimate *ex ante*.

Table 10: Fitted deviation constants at coverage $q = 0.90$ (train-line-weighted, Section 4.5), and the as-is coverage of the planner-available M3 (5-block train-internal std). c_1 : M1 proportional; c_2 : M2 square-root. Source: `results/gamma/gamma_fits.csv`.

Cut	iscf480 temporal			iscf10kt temporal		
	$c_1(0.9)$	$c_2(0.9)$	M3 cov. (%)	$c_1(0.9)$	$c_2(0.9)$	M3 cov. (%)
r0f0	0.76	46.2	5.4	1.31	5.5	79.5
r0f1	0.94	66.3	3.4	1.42	6.6	76.1
r0f2	1.18	96.5	4.2	1.63	9.7	72.0
r0f3	1.55	146.9	1.5	2.45	13.5	58.0
r0f4	1.48	210.8	1.7	3.20	22.8	37.6

Table 11: Budget diagnostics at the $q = 0.90$ fit, $k=1$. Γ^{cover} over the stations violating the normalized check (iscf480: $n = 14$ station-cells of 60 products each; iscf10kt: $n = 11$ of 541); Γ^{bind} over all 40 station-cells per family. “ ∞ ” = uncoverable even at $\Gamma = |\Phi_s|$ (full box) under the model’s magnitudes: for M1/M2 driven by train-unseen products ($\hat{L}_p = 0$), for M3 by under-sized block-std magnitudes. Source: `results/gamma/gamma_station.csv`.

Family	Model	Γ^{cover} (violated stations)			Γ^{bind} (all stations)		
		$\#\infty$	median	median (finite)	median	% = 1	% ≤ 2
iscf480	M1 prop.	1	4.5	4	1	65.0	80.0
iscf480	M2 sqrt	0	5	5	1	67.5	80.0
iscf480	M3 empirical	10	∞	31	3	40.0	47.5
iscf10kt	M1 prop.	0	8	8	1	67.5	75.0
iscf10kt	M2 sqrt	0	28	28	1	57.5	70.0
iscf10kt	M3 empirical	3	135	77.5	1	57.5	60.0

Budgets (Table 11, Figure 5). On the stations that actually broke the as-run check, the best-fitting model needs remarkably small budgets: under M2 on iscf480 the seven as-run-broken stations need $\Gamma^{\text{cover}} \in \{0, 1, 2, 8, 20, 23, 24\}$, median 8 of 60 assigned products (13.3%; M2 is the best mean there, 11.1, with all covers finite, whereas M1 leaves 1/7 uncoverable and M3 4/7); under M1 on iscf10kt they need $\{5, 19, 27\}$, median 19 of 541 (3.5%, range 0.9–5.0%). Over all normalized-violated stations the medians remain small relative to ζ_s (Table 11). At the same time $\Gamma^{\text{bind}} = 1$ for 58–68% of $k=1$ stations under M1/M2 (median 1; ≤ 2 for 70–80%): with these fitted magnitudes, protecting even the single largest deviation already exceeds the 10%-slack train ceiling at that majority of stations, so there *any* nonzero budget forces the optimizer to rebalance at train time (at the remaining stations Γ^{bind} reaches 109 under M1, so small budgets exert no pressure). Under M1/M2 the robust constraint would be vacuous nowhere (Γ^{bind} finite in all 80 $k=1$ station-cells each); under M3, 12 of 80 station-cells tolerate any budget ($\Gamma^{\text{bind}} = \infty$).

Concentration (Figure 6). Overload is head-concentrated but not degenerate: across the 25 violating (fold, station) cells at $k=1$, the median Gini of positive deviations is 0.63 (iscf480) and 0.82 (iscf10kt); the top-1 product covers 15–50% (iscf480) and 16–53% (iscf10kt) of a broken station’s total positive deviation, and the top-10 covers 69–99% and 61–89%. A moderate head of products, not one or two outliers and not diffuse noise, drives the overloads — exactly the small- Γ regime in which the budget model of Section 4 is designed to operate.

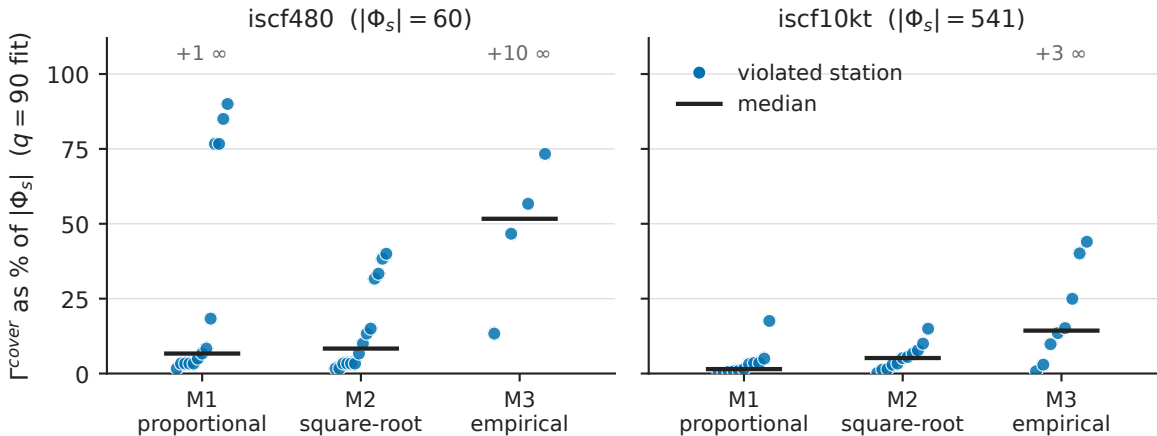


Figure 5: Γ^{cover} at the $q = 0.90$ fit for $k=1$ stations violating the normalized check, as a fraction of assigned products: iscf480 medians $\approx 7\%$ (M1), 8% (M2), 52% (M3) of $|\Phi_s| = 60$, with 1 (M1) and 10 (M3) infinite covers annotated; iscf10kt medians $\approx 1.5\%$ (M1), 5% (M2), 14% (M3) of 541, with 3 M3 infinities. Source: `results/gamma/gamma_station.csv`.

4 A budget-of-uncertainty robust counterpart for the workload constraint

The empirical companion (Section 3) documents a failure mode of trained CSLAP layouts under demand drift: layouts that satisfy the station workload constraint on the training window violate it out-of-sample on early temporal cuts. This section develops the exact robust counterpart of the workload constraint under budgeted pick-line uncertainty, targeting this failure mode; how much protection is actually delivered depends on the calibrated uncertainty set and is assessed

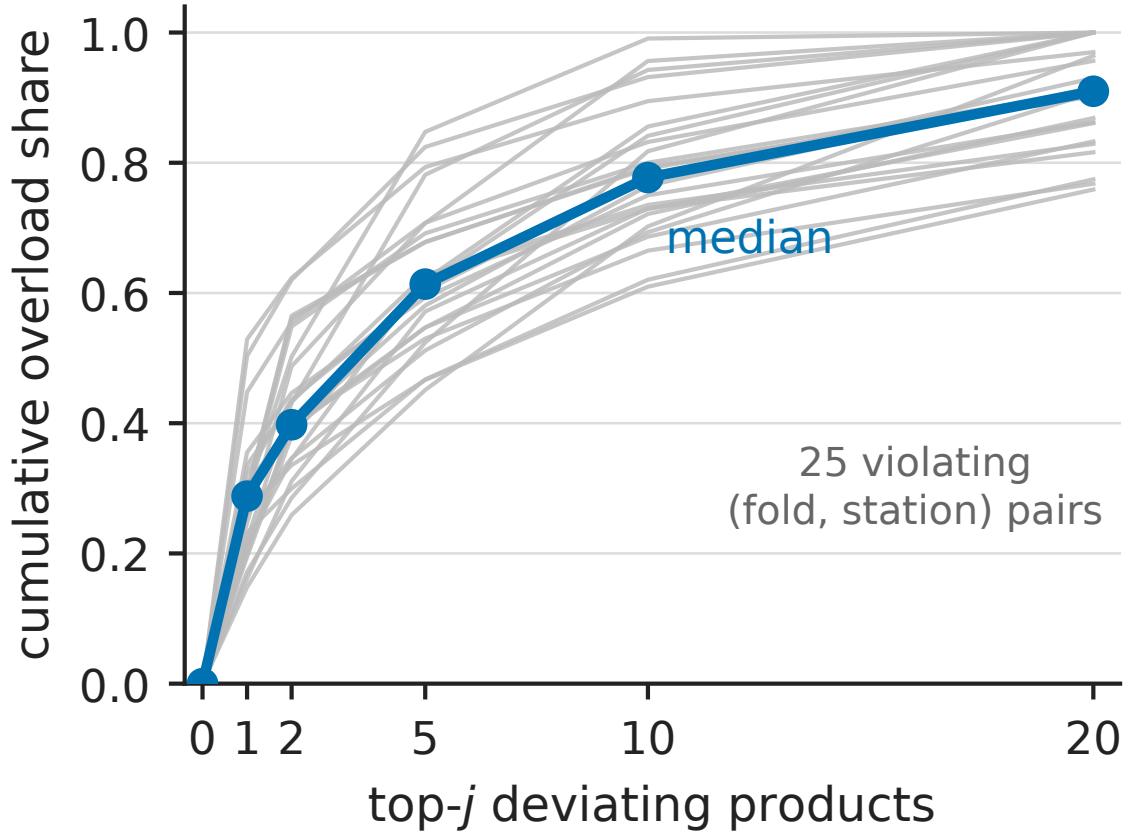


Figure 6: Cumulative share of a broken station’s total positive deviation covered by its top- m deviating products, for the 25 violating (fold, station) cells at $k=1$ (both temporal families). Median cumulative share ≈ 0.29 at top-1, ≈ 0.61 at top-5, ≈ 0.78 at top-10. Source: `results/gamma/concentration.csv`.

in Section 4.5. Uncertainty is placed on the pick-line coefficients L_p only; the visits objective is untouched. The construction is the budget-of-uncertainty model of [1], in the row-wise robust-optimization framework of [4], specialized to the CSLAP placement MILP. Symbols follow the project notation: P , O , S are the product, order, and station sets; P_o the support of order o ; ζ_s , V_s , T_s the slot capacity, processing speed, and time capacity of station s .

4.1 Nominal placement model

The nominal placement model assigns every product to exactly one station and minimizes the total number of station visits over the training orders. The decision variables are $x_{ps} \in \{0, 1\}$ (product $p \in P$ assigned to station $s \in S$) and the auxiliary visit variables $z_{os} \in [0, 1]$ (order $o \in O$ visits station s , completely dependent on x_{ps}):

$$\min \sum_{o \in O} \sum_{s \in S} z_{os} \quad (3)$$

subject to

$$\sum_{s \in S} x_{ps} = 1 \quad \forall p \in P, \quad (4)$$

$$\sum_{p \in P} x_{ps} \leq \zeta_s \quad \forall s \in S, \quad (5)$$

$$x_{ps} \leq z_{os} \quad \forall p \in P_o, \forall o \in O, \forall s \in S, \quad (6)$$

$$\sum_{p \in P} x_{ps} \cdot \frac{L_p}{V_s} \leq T_s \quad \forall s \in S, \quad (7)$$

$$x_{ps} \in \{0, 1\}, \quad z_{os} \in [0, 1], \quad \forall p \in P, o \in O, s \in S.$$

Constraint (4) assigns each product to exactly one station, (5) is the physical slot capacity, (6) links assignments to visits (a logical OR; no big- M is needed because the objective pushes z_{os} down to $\max_{p \in P_o} x_{ps}$), and (7) bounds the workload of each station over the planning window.⁵

The workload coefficient L_p is a *point estimate*: the training-window pick-line count of product p , computed as the number of distinct training orders containing p (`REAL_LINES`; on instances with duplicate order-product rows the raw-line and distinct-order conventions differ — see the BERNER caveat in Section 3). The time capacity is calibrated on the same window by the project’s 10% operational-slack rule,

$$T_s = \left\lceil 1.10 \cdot \frac{\sum_{p \in P} L_p}{V_s |S|} \right\rceil, \quad (8)$$

so a layout solving (3)–(7) is guaranteed workload-feasible *only for the estimate L_p of the training window*. When the realized pick-line vector of a later operating window drifts away from the estimate — as the temporal experiments of Section 3 show it does — (7) offers no protection: a single product whose demand doubles can push its station past T_s while the model still certifies feasibility.

⁵The manuscripts typeset the summation in (7) as $\sum_{p \in P_o}$ with no quantifier on o ; the column-generation pricing model and the implementation both sum over *all* products assigned to the station, so we write $\sum_{p \in P}$ here.

4.2 Budget-of-uncertainty model for pick-line workloads

Let L_p^0 denote the nominal (training-window) estimate of the pick-line count — the quantity written L_p in (7) — and let $\hat{L}_p \geq 0$ be a deviation magnitude for product p .⁶ The uncertain coefficient is modeled as

$$\tilde{L}_p \in [L_p^0 - \hat{L}_p, L_p^0 + \hat{L}_p] \quad \forall p \in P, \quad (9)$$

with the budget restriction of [1]: in any realization, *at most* Γ products deviate from their nominal value simultaneously, where $\Gamma \in \{0, 1, \dots, |P|\}$ is the conservatism knob. Since the workload constraint is a $\leq$ row with nonnegative coefficients, only upward deviations threaten it, so the worst case places every deviating coefficient at its upper endpoint $L_p^0 + \hat{L}_p$.

Write $\Phi_s = \{p \in P : x_{ps} = 1\}$ for the set of products assigned to station s (the Hexaly set-variable of the project notation). Requiring (7) to hold for every realization in the budgeted set gives the robust workload constraint

$$\sum_{p \in \Phi_s} \frac{L_p^0}{V_s} x_{ps} + \max_{\substack{J \subseteq \Phi_s \\ |J| \leq \Gamma}} \sum_{p \in J} \frac{\hat{L}_p}{V_s} x_{ps} \leq T_s \quad \forall s \in S. \quad (10)$$

On Φ_s the factors $x_{ps} = 1$ are redundant; they are retained to make the decision-dependence of the adversary's ground set explicit, and they let the sums range over all of P without changing value (terms with $x_{ps} = 0$ vanish). The inner combinatorial maximization admits the equivalent continuous relaxation

$$\psi_s(x; \Gamma) = \max_z \sum_{p \in P} \frac{\hat{L}_p}{V_s} x_{ps} z_p \quad \text{s.t.} \quad \sum_{p \in P} z_p \leq \Gamma, \quad 0 \leq z_p \leq 1 \quad \forall p \in P. \quad (11)$$

For fixed x , (11) is a linear program in z alone: the x enter only through the constant objective coefficients $d_{ps} := (\hat{L}_p/V_s) x_{ps} \geq 0$. Its constraint matrix — one cardinality row plus the identity box — is an interval matrix, hence totally unimodular; equivalently, by the greedy argument, since all $d_{ps} \geq 0$ an optimal z sets $z_p = 1$ on the Γ largest coefficients (on all positive ones if fewer than Γ) and $z_p = 0$ elsewhere. For integer Γ integrality of the inner adversarial problem is therefore *free*, and $\psi_s(x; \Gamma)$ equals the set-maximum in (10).⁷

Proposition 1 (Exact linear counterpart). *Fix a station $s \in S$, an integer budget $\Gamma \in \mathbb{Z}_{\geq 0}$, and a binary assignment vector $x_s \in \{0, 1\}^P$.⁸ The robust workload constraint (10) holds if and only if there exist multipliers $\theta_s \geq 0$ and $\mu_{ps} \geq 0$, $p \in P$, such that*

$$\sum_{p \in P} \frac{L_p^0}{V_s} x_{ps} + \Gamma \theta_s + \sum_{p \in P} \mu_{ps} \leq T_s, \quad (12)$$

$$\theta_s + \mu_{ps} \geq \frac{\hat{L}_p}{V_s} x_{ps} \quad \forall p \in P. \quad (13)$$

⁶The superscripted symbol L_p^0 is chosen deliberately: the accented $\bar{L}_p$ is reserved by the covering draft for the closure-inflated line count (and $\bar{T}_s$ for its recalibrated capacity), so no renaming rule is needed when the two studies are merged into one manuscript.

⁷The adversary variable z_p (single subscript) is unrelated to the visit variable z_{os} and to the cover-pricing z_p of the covering draft; it exists only inside (11) and is dualized away in Proposition 1.

⁸Integrality of Γ is what makes the LP value $\psi_s(x; \Gamma)$ of (11) coincide with the set-maximum in (10); for fractional Γ the $(\Rightarrow)$ direction fails: one assigned product with $\hat{L}_p/V_s = 1$, zero nominal slack, and $\Gamma = 1/2$ satisfies (10) (only $J = \emptyset$ is admissible), yet $\psi_s(x; 1/2) = 1/2 > 0$, so no multipliers exist. Real-valued budgets can be recovered by adding the Bertsimas–Sim fractional deviation term $(\lfloor \Gamma \rfloor \text{ full deviations plus one scaled by } \Gamma - \lfloor \Gamma \rfloor)$ to (10); integer budgets suffice here, consistent with $\Gamma \in \{0, 1, \dots, |P|\}$ above.

Since $\hat{L}_p/V_s$ and L_p^0/V_s are data constants, each right-hand side is a constant multiple of the decision variable x_{ps} , so (12) and (13) are linear in (x, θ, μ) ; no integrality of x is needed for linearity.

Proof. By the discussion following (11) — the step in which integrality of Γ is used — constraint (10) is equivalent to

$$\psi_s(x; \Gamma) \leq T_s - \sum_{p \in P} \frac{L_p^0}{V_s} x_{ps} =: \rho_s(x), \quad (14)$$

where $\rho_s(x)$ is the nominal slack of station s . The inner LP (11), with coefficients $d_{ps} = (\hat{L}_p/V_s) x_{ps} \geq 0$, is *feasible* ($z = 0$) and *bounded above* ($\sum_p d_{ps} z_p \leq \sum_p d_{ps} < \infty$ since $z_p \leq 1$), so it attains a finite optimum $\psi_s(x; \Gamma)$. Associate the dual variable $\theta_s \geq 0$ with the budget row $\sum_p z_p \leq \Gamma$ and $\mu_{ps} \geq 0$ with each upper bound $z_p \leq 1$. The LP dual of (11) is

$$\min_{\theta_s, \mu_{ps}} \Gamma \theta_s + \sum_{p \in P} \mu_{ps} \quad \text{s.t.} \quad \theta_s + \mu_{ps} \geq d_{ps} \quad \forall p \in P, \quad \theta_s \geq 0, \mu_{ps} \geq 0. \quad (15)$$

Because the primal is feasible and bounded, LP strong duality applies: the dual (15) is solvable and its optimal value equals $\psi_s(x; \Gamma)$. We prove the two implications separately.

($\Leftarrow$) Suppose $(\theta_s, \mu_{ps}) \geq 0$ satisfies (12)–(13). Constraint (13) says the point is feasible for the dual (15), so by *weak* duality $\psi_s(x; \Gamma) \leq \Gamma \theta_s + \sum_p \mu_{ps}$. Rearranging (12) gives $\Gamma \theta_s + \sum_p \mu_{ps} \leq \rho_s(x)$. Chaining the two inequalities yields $\psi_s(x; \Gamma) \leq \rho_s(x)$, which is (14), i.e., the robust constraint (10) holds.

($\Rightarrow$) Conversely, suppose (10) holds, i.e., $\psi_s(x; \Gamma) \leq \rho_s(x)$. By strong duality the dual (15) attains its optimum at some $(\theta_s^*, \mu_{ps}^*) \geq 0$ with

$$\Gamma \theta_s^* + \sum_{p \in P} \mu_{ps}^* = \psi_s(x; \Gamma) \leq \rho_s(x) = T_s - \sum_{p \in P} \frac{L_p^0}{V_s} x_{ps}.$$

The optimal dual point is dual-feasible, so it satisfies (13); the displayed inequality is exactly (12). Hence the required multipliers exist.

Finally, linearity: $\hat{L}_p/V_s$ and L_p^0/V_s are constants, so (12) is linear in (x, θ, μ) and the right-hand side of (13) is a constant multiple of the decision variable x_{ps} — no product of decision variables appears anywhere, whether or not x is integral. $\square$

4.3 The robust placement MILP

Applying Proposition 1 station by station — with a per-station integer budget $\Gamma_s \in \mathbb{Z}_{\geq 0}$, the uniform choice $\Gamma_s = \Gamma$ being the default — and substituting the counterpart for (7) yields the full

robust placement MILP:

$$\min_{x, z, \theta, \mu} \sum_{o \in O} \sum_{s \in S} z_{os} \quad (16a)$$

$$\text{s.t.} \quad \sum_{s \in S} x_{ps} = 1 \quad \forall p \in P, \quad (16b)$$

$$\sum_{p \in P} x_{ps} \leq \zeta_s \quad \forall s \in S, \quad (16c)$$

$$x_{ps} \leq z_{os} \quad \forall p \in P, \forall o \in O, \forall s \in S, \quad (16d)$$

$$\sum_{p \in P} \frac{L_p^0}{V_s} x_{ps} + \Gamma_s \theta_s + \sum_{p \in P} \mu_{ps} \leq T_s \quad \forall s \in S, \quad (16e)$$

$$\theta_s + \mu_{ps} \geq \frac{\hat{L}_p}{V_s} x_{ps} \quad \forall p \in P, \forall s \in S, \quad (16f)$$

$$x_{ps} \in \{0, 1\}, \quad z_{os} \in [0, 1], \quad \theta_s \geq 0, \quad \mu_{ps} \geq 0 \quad \forall p, o, s. \quad (16g)$$

The objective (16a) and the combinatorial core (16b)–(16d) are identical to the nominal model (3), (4)–(6); only the workload block changes. The size overhead over the nominal MILP is exactly $+|S|$ continuous variables θ_s , $+|P||S|$ continuous variables μ_{ps} , and $+|P||S|$ dual-feasibility constraints (16f); the workload rows stay at $|S|$, the binaries and the objective are unchanged. Robustness is bought with one continuous LP block, not with additional integer structure.

4.4 Properties

Proposition 2 (Budget limits and monotonicity). *For an integer budget $\Gamma \in \mathbb{Z}_{\geq 0}$, let*

$$\mathcal{A}(\Gamma) = \{x \in \{0, 1\}^{P \times S} : \exists (z, \theta, \mu) \text{ such that } (x, z, \theta, \mu) \text{ is feasible for (16) with } \Gamma_s = \Gamma\}$$

denote the projection onto the assignment variables of the feasible set of (16) with uniform budget Γ . Then:

- (i) $\Gamma = 0$ recovers the nominal model: $\mathcal{A}(0)$ equals the projection onto x of the feasible set of (4)–(7) with $L_p = L_p^0$.
- (ii) If $\Gamma \geq \zeta_s$ for every s , then (10) reduces to the box (Soyster) counterpart $\sum_{p \in P} \frac{L_p^0 + \hat{L}_p}{V_s} x_{ps} \leq T_s$ [3].
- (iii) $\Gamma \mapsto \mathcal{A}(\Gamma)$ is nonincreasing: $\Gamma' \geq \Gamma$ implies $\mathcal{A}(\Gamma') \subseteq \mathcal{A}(\Gamma)$, and the optimal visit count of (16) is nondecreasing in Γ .

Proof. (i) Both projections are cut out by the common core (4)–(6); the visit variables never restrict x further (setting $z_{os} = 1$ is always feasible), so it suffices to compare the workload blocks. With $\Gamma = 0$ the inner maximum in (10) is over $J = \emptyset$ only and equals 0, so (10) is (7). In the counterpart, given x satisfying (7), choose $\theta_s = \max_p \hat{L}_p / V_s$ and $\mu_{ps} = 0$: (13) holds and $\Gamma_s \theta_s$ vanishes, so (12) is exactly (7); conversely (12) with $\Gamma_s = 0$ and $\mu \geq 0$ implies (7).

(ii) By the slot capacity (5), $|\Phi_s| = \sum_p x_{ps} \leq \zeta_s \leq \Gamma$, so the restriction $|J| \leq \Gamma$ is vacuous and the adversary deviates *every* assigned product upward: the inner maximum is $\sum_{p \in \Phi_s} \hat{L}_p / V_s$, the worst case of the full interval box (9). Any $\Gamma \geq \max_s |\Phi_s|$ suffices; since $|\Phi_s|$ is decision-dependent, $\Gamma \geq \zeta_s$ is the natural data-level sufficient condition.

(iii) $\psi_s(x; \Gamma)$ is nondecreasing in Γ (the feasible set of (11) grows with Γ), so the feasible sets $\mathcal{A}(\Gamma)$ are nested and minimizing the same objective over a smaller set cannot decrease the optimum. $\square$

Thus Γ interpolates monotonically between the nominal model and full box conservatism, exactly the Bertsimas–Sim design.

Remark 1 (Decision-dependent adversary support). *The adversary’s ground set Φ_s in (10) is itself a decision, since $\Phi_s = \{p : x_{ps} = 1\}$. Endogenous uncertainty sets are in general a hard extension of robust optimization, but no extra machinery is needed here: because the factor x_{ps} is kept inside the inner objective of (11), deviating a non-assigned product contributes zero, so the adversary restricted to Φ_s and the adversary ranging over all of P attain the same value. The dualization of Proposition 1 then applies verbatim. Linearity of the counterpart is automatic because $\hat{L}_p/V_s$ are data constants (no bilinear terms, no big- M); what binariness of x_{ps} buys is this ground-set equivalence, not linearity. (Under merely nonnegative x_{ps} the equivalence holds with respect to the support $\{p : x_{ps} > 0\}$ rather than Φ_s : a fractional $0 < x_{ps} < 1$ lies outside Φ_s yet contributes $(\hat{L}_p/V_s)x_{ps} > 0$ to (11).)*

Remark 2 (Probabilistic guarantee). *Suppose the realized deviations of the assigned products are independent and symmetrically distributed on the intervals (9). Then, for any assignment x satisfying the protected row (10) with budget Γ — in particular, any x feasible for (16) — [1] bound the violation probability of that row by*

$$\Pr\left[\sum_{p \in \Phi_s} \tilde{L}_p/V_s > T_s\right] \leq B(n, \Gamma) \leq \exp(-\Gamma^2/(2n)), \quad n = |\Phi_s| \leq \zeta_s, \quad (17)$$

where $B(n, \Gamma)$ is the exact binomial bound of Theorem 2 of [1], well approximated by $1 - \Phi_N((\Gamma - 1)/\sqrt{n})$ with Φ_N the standard normal c.d.f. (not the assigned set Φ_s); the closed-form exponential expression is the weaker Hoeffding-type bound of [2], derived for homogeneous deviation magnitudes across the row. A budget of order $\sqrt{n}$ already yields a strong guarantee at a small fraction of the box protection level. One caveat: daily pick-line counts are nonnegative and right-skewed, so the symmetric-deviation assumption behind (17) is an approximation here; the bound is indicative, and in practice it is replaced by the data-driven procedure of Section 4.5: deviation magnitudes fitted on train-internal blocks, a budget fixed *ex ante* on a grid, and retrospective diagnostics assessing adequacy.

Remark 3 (Constraint-side uncertainty only; relation to H1). *Uncertainty enters model (16) through the constraints only; the visits objective (16a) is identical to the nominal one. This is deliberately complementary to the falsified covering hypothesis H1, which perturbed the objective by enlarging every order to its pattern-closure and paid a certified out-of-sample price ($G(\kappa) < 0$ at $\kappa = 2$ on the clean instance). Here nothing is insured about the visit count: the insured quantity is workload-feasibility risk — the out-of-sample violations $W_s^n > T_s$ documented in Section 3 — which the nominal model cannot see by construction. The two mechanisms are orthogonal; the present study evaluates the workload channel alone.*

Remark 4 (Volume basis and reporting convention). *The counterpart is meaningful only if L_p^0 , $\hat{L}_p$, and T_s are expressed on the same volume basis. In the project’s adapters, T_s derives from the training-window totals by the 10% slack rule (8) (κ -invariant on the clean instance); accordingly, L_p^0 must be the training-window line count and $\hat{L}_p$ a deviation of volume-normalized counts on that basis (Section 4.5). Mixing bases — e.g., a test-window $\hat{L}_p$ against a train-calibrated T_s — would re-introduce the recalibration confound the clean instance was built to remove. Since the protection level depends on $n = |\Phi_s| \leq \zeta_s$, we recommend reporting budgets as the dimensionless fraction Γ_s/ζ_s , comparable across stations and instances.*

4.5 Data-driven calibration of $(\hat{L}_p, \Gamma)$

The counterpart of Section 4.3 is fully specified once the deviation magnitudes $\hat{L}_p$ and the budget Γ are chosen. We define the estimators the empirical section instantiates; no implementation is discussed.

Deviation models. Let $\delta_p^{(b)} = L_p^{(b)} - L_p^0$ denote the realized deviation of the volume-normalized line count $L_p^{(b)}$ of product p on a train-internal temporal block $b \in \{1, \dots, B\}$ (contiguous segments of the training window by order rank; counts are rescaled so each block carries the training-window total volume, keeping everything on the basis of Remark 4). The three deviation models are

$$\text{M1: } \hat{L}_p = c_1 L_p^0, \quad \text{M2: } \hat{L}_p = c_2 \sqrt{L_p^0}, \quad \text{M3: } \hat{L}_p = \text{std}_{b=1, \dots, B}(L_p^{(b)}). \quad (18)$$

M1 assumes deviations proportional to demand (multiplicative drift), M2 assumes the $\sqrt{L_p^0}$ scaling of count noise (Poisson-like), and M3 is the nonparametric per-product standard deviation across the training blocks. M3 needs no fitted constant; M1 and M2 are calibrated as follows.

Fitting c . For a coverage target $q \in (0, 1)$ (e.g. $q = 0.9$), the constant is the smallest value whose implied intervals cover a L^0 -weighted fraction q of the realized deviations on the train-internal blocks:

$$c(q) = \min \left\{ c \geq 0 : \sum_{p, b} L_p^0 \mathbf{1}[|\delta_p^{(b)}| \leq c g(L_p^0)] \geq q \sum_{p, b} L_p^0 \right\}, \quad (19)$$

with $g(L_p^0) = L_p^0$ for M1 and $g(L_p^0) = \sqrt{L_p^0}$ for M2; equivalently, $c(q)$ is the L^0 -weighted q -quantile of the normalized deviations $|\delta_p^{(b)}|/g(L_p^0)$. Weighting by L_p^0 prioritizes high-volume products, whose deviations dominate station workloads; restricting to train-internal blocks keeps the calibration free of test leakage.

Budget diagnostics. Fix a trained layout a (indicator x^a), a station s with assigned set Φ_s , nominal load $\bar{W}_s = \sum_{p \in \Phi_s} L_p^0/V_s$, and realized normalized load $W_s^n = \sum_{p \in \Phi_s} L_p^n/V_s$, where L_p^n is the volume-normalized realized line count of the evaluation window. By (11), $\psi_s(x^a; \Gamma) = \frac{1}{V_s} \sum_{p \in J_\Gamma(s)} \hat{L}_p$ with $J_\Gamma(s) \subseteq \Phi_s$ the $\min(\Gamma, |\Phi_s|)$ largest deviations at station s . Two *retrospective* diagnostics summarize how large a budget the data ask for and how large a budget the training instance tolerates:

$$\Gamma^{\text{cover}}(s) = \min \left\{ \Gamma \in \mathbb{Z}_{\geq 0} : \bar{W}_s + \psi_s(x^a; \Gamma) \geq W_s^n \right\}, \quad (20)$$

$$\Gamma^{\text{bind}}(s) = \min \left\{ \Gamma \in \mathbb{Z}_{\geq 0} : \bar{W}_s + \psi_s(x^a; \Gamma) > T_s \right\}. \quad (21)$$

$\Gamma^{\text{cover}}(s)$ is the smallest budget whose protection level reaches the realized load of station s : the budget that would have been needed for the robust constraint to “see” the observed drift (it is 0 when $\bar{W}_s \geq W_s^n$, and undefined — reported as ∞ — when even $\Gamma = |\Phi_s|$ under-covers, i.e., when the deviation model itself is too small). $\Gamma^{\text{bind}}(s)$ is the smallest budget at which the robust left-hand side exceeds T_s on the training data, i.e., at which the counterpart starts to cut off the trained layout; budgets below it exert no retraining pressure at station s , budgets at or above it force the optimizer to rebalance. Symmetrically to Γ^{cover} , the set in (21) may be empty: when even the full box load keeps the station within capacity, $\bar{W}_s + \sum_{p \in \Phi_s} \hat{L}_p/V_s \leq T_s$, no finite budget makes the constraint bind (ψ_s saturates at $\Gamma = |\Phi_s|$), and we set $\Gamma^{\text{bind}}(s) = \infty$ — such stations tolerate any budget.

Use in the empirical study. In a future implementation phase the operative budget would be fixed *ex ante*: robust layouts would be trained on a predeclared grid of budgets (reported as fractions of ζ_s per Remark 4), with the deviation magnitudes fitted on train-internal blocks only via (19), so that no evaluation-window quantity would enter the choice of Γ for any scored test cut. No robust layout is trained in this study: Section 3 computes only the *retrospective* diagnostics (20)–(21) on the existing nominal layouts — for each temporal cut and deviation model in (18), the distributions of $\Gamma^{\text{cover}}(s)$ over the stations that actually violate $W_s^n > T_s$ out-of-sample and of $\Gamma^{\text{bind}}(s)$ over *all* stations (healthy and violated alike) — answering “what budget *would have been* needed” and “what budget the training instance tolerates”. The empirical instantiation also departs from (19) on the magnitude side: the reported constants c_1, c_2 (Section 3.8) apply the same L^0 -weighted quantile formula *retrospectively* to the realized evaluation-window deviations $\delta_p = L_p^n - L_p^0$ — they are diagnostic, not planner-available — and only M3 uses train-internal blocks as (19) prescribes. A budget Γ would be judged *adequate in retrospect* when it dominates the observed Γ^{cover} quantiles on drift-hit stations — so the protected constraint would have flagged the realized violations, and the robust model is designed to rebalance them away at train time — while staying below Γ^{bind} on most healthy stations, so protection is bought where drift occurs rather than by uniform slack. Once an ex-ante grid exists, these curves would audit it after the fact rather than select the budget: they replace the a priori appeal to the symmetric-noise bound (17) as the lens for *judging* Γ , and they must not feed back into its choice on any window used for scoring.

5 Computational study: implementing the robust counterpart

The preceding section was written, in the first version of this report, as a theory-only contribution. This section removes that restriction: it implements the robust placement MILP of Equation (16) and evaluates it under the same temporal protocol as Section 3, on iscf480. Three questions drive the design. First, is the counterpart *implementable at all* under the project’s own capacity convention — the $\sigma = 1.10$ slack rule that calibrates T_s ? Second, when it is implementable, what does the protection buy out of sample, and at what price in visits? Third, does *targeted* budgeted protection beat the obvious nominal alternative — uniformly tightening the training ceiling by the same amount of visit price?

5.1 Setup

Solver. No commercial solver is available on the experiment machine, so Equation (16) is implemented on the open-source HiGHS solver (`src/milp_highs_robust.py`, via `highspy`). The visit variables are declared continuous, which is exact: for fixed integral x_{ps} the minimal $y_{os} = \max_{p \in P_o} x_{ps}$ is automatically binary, so the model carries only the $|P||S| = 3,840$ assignment binaries plus the continuous dual block of Proposition 1. Exact branch-and-bound is nonetheless out of reach at useful time scales: the eight stations of iscf480 are identical, and the resulting symmetry left HiGHS unable to improve — or even find — an incumbent within minutes. Incumbents therefore come from a three-stage pipeline: (i) a feasibility-first greedy (best-fit decreasing on the *robust* station load, swap repair); (ii) a deterministic pairwise-swap local search on the weighted visit objective, with the *original* robust constraint — nominal load plus the sum of the Γ largest deviations, per station — as a hard accept filter (three seeded restarts, pass-budgeted for cross-process reproducibility); (iii) a warm-started HiGHS run used for what an exact solver is still good for here: *proofs of infeasibility* for budgets that do not fit, and dual bounds. The optimization objective aggregates the top 1,500 weighted distinct multi-item supports of the training orders; all reported visit counts, however, are evaluated on the *full real* order sets, and all feasibility metrics

on the fixed 1.10-rule contract $T^n = \lceil 1.10 \sum_p L_p^{te} / (V|S|) \rceil$ of Section 3. Every solved robust layout was re-verified against the pre-dualization constraint by direct top- Γ arithmetic (`robust_ok` in the result tables): 21 of 21 feasible robust arms pass, so the dual block never “cheats.”

Deviations: ex-ante M3 with partial insurance. The deviations $\hat{L}_p$ are calibrated with the only model of Section 4.5 available at decision time: M3, the per-product ddof-1 standard deviation of the five volume-rescaled temporal train-block line counts, capped at $\bar{L}_p$. (The global- c M2 fit degenerates here: temporally concentrated SKUs push the $\bar{L}$ -weighted quantile to $c \approx 70$ –97, at which point the cap binds for essentially the whole catalogue and the set collapses to the Soyster box.) Because Section 3.8 showed M3 *underestimates* realized drift on iscf480, and because — as Section 5.2 shows — even M3-sized deviations do not fit the slack contract in full, the experiment adds a partial-insurance factor $\alpha \in \{0.5, 0.25\}$: the robust constraint is built with $\alpha \hat{L}_p$ in place of $\hat{L}_p$. Formally this is just Equation (16) on a shrunk uncertainty set, so Proposition 1 applies verbatim; operationally it asks what protection is *purchasable* inside the existing capacity margin.

Arms and pairing. For each of the five expanding temporal cuts of Section 3: the nominal arm $\Gamma = 0$; budgeted arms $\Gamma \in \{1, 2, 4, 8\} \times \alpha \in \{0.5, 0.25\}$; full- σ probes $\Gamma \in \{1, 4, 8, 16\}$ at $\alpha = 1$ (cut 0); two *uniform-tightening* competitors, trained with the ceiling $T_s(\beta) = \lceil \beta \sum_p \bar{L}_p / (V|S|) \rceil$ for $\beta \in \{1.05, 1.02\}$ and $\Gamma = 0$ — the same visit price paid as undirected balance instead of directed protection; and the stored Hexaly $k=1$ layout as an external reference. Every arm of a cut shares the same supports, budgets, calibration and evaluation, so within-cut comparisons are paired; the price of robustness $\text{PoR} = (V_{tr}(\cdot) - V_{tr}(\Gamma=0)) / V_{tr}(\Gamma=0)$ and the test-visit gain $G = (V_{te}(\Gamma=0) - V_{te}(\cdot)) / V_{te}(\Gamma=0)$ are measured against the paired nominal arm.

5.2 Full- σ protection does not fit the slack contract

The 1.10 rule leaves each station a nominal headroom of $(1 - 1/1.10) T_s$ — about 4,124 lines per station on cut 0 ($T_s = 45,365$). The M3 deviations are far larger than that: the top SKU alone carries $\hat{L}_p = 13,307$ (its block- σ exceeds its own nominal, so the cap binds), and the eight largest deviations sum to 48,621 — one and a half times the *total* headroom $8 \times 4,124 \approx 33,000$ of the whole cell. Since the heavy deviators are also heavy products, the nominal constraint prevents parking them all in one station, and full- σ protection has, arithmetically, nowhere to live. HiGHS makes this formal at moderate budgets: at $\alpha = 1$ it returns a **proven infeasibility** certificate for $\Gamma = 4, 8$ and 16 on cut 0, within a minute per arm (28–41 s of total arm time including the constructive pipeline). The $\Gamma = 1$ case is harder — its LP relaxation is feasible — and remains **undetermined**: no incumbent and no certificate after a 318-second probe and a separate 820-second attempt (solver logs). The same caution applies at $\alpha < 1$: the arms marked infeasible in Table 12 ($\Gamma \geq 4$ at $\alpha = 0.5$, $\Gamma = 8$ at both α , and $\Gamma = 2$ at $\alpha = 0.5$ on four of five cuts) are *undetermined-within-budget* — neither the constructive pipeline nor a 180-second HiGHS probe produced a layout — not proven infeasible. At half- σ , by contrast, the top-eight deviation mass is $24,310 < 33,000$, and indeed $\Gamma = 1, \alpha = 0.5$ is constructively feasible on all five cuts.

The practical reading is blunt: under the project’s own capacity convention, the drift that iscf480 actually exhibits (Section 3.8) is too large to be insured in full. Either the protection target is scaled down (α), or the slack σ — a capacity investment — must grow. The remainder of the section quantifies the first option.

Table 12: Per-arm means over the temporal cuts on which the arm is feasible (iscf480; evaluation on real test orders against the fixed 1.10 contract T^n). PoR and G are paired against the $\Gamma=0$ arm of the same cut; $G < 0$ means the arm pays visits out of sample too. Arms absent from the body: at $\alpha = 1$, $\Gamma \in \{4, 8, 16\}$ *proven* infeasible (HiGHS certificates, cut 0) and $\Gamma = 1$ undetermined; at $\alpha = 0.5$, $\Gamma = 4, 8$ undetermined on 5/5 cuts; at $\alpha = 0.25$, $\Gamma = 8$ undetermined on 5/5 cuts.

arm	cuts	PoR %	G %	viol. (/8)	peak W_s/T^n	worst peak	X %
$\Gamma=0$ (nominal)	5/5	0.00	0.00	2.6	1.77	2.11	17.4
$\Gamma=1$, $\alpha=0.5$	5/5	2.34	-4.44	2.6	1.59	1.87	13.0
$\Gamma=1$, $\alpha=0.25$	5/5	1.44	-2.49	2.6	1.67	1.90	15.4
$\Gamma=2$, $\alpha=0.25$	5/5	1.41	-1.64	2.2	1.73	1.94	15.5
$\Gamma=4$, $\alpha=0.25$	5/5	1.77	-2.41	2.8	1.74	2.11	15.2
$\Gamma=2$, $\alpha=0.5$	1/5	19.13	-23.14	3.0	1.14	1.14	3.1
tightened $\beta=1.05$	5/5	0.41	-0.28	2.8	1.76	2.03	16.8
tightened $\beta=1.02$	5/5	2.26	-1.73	2.8	1.75	2.01	16.9
Hexaly $k=1$ (reference)	5/5	-3.88	6.16	2.8	2.15	2.52	23.1

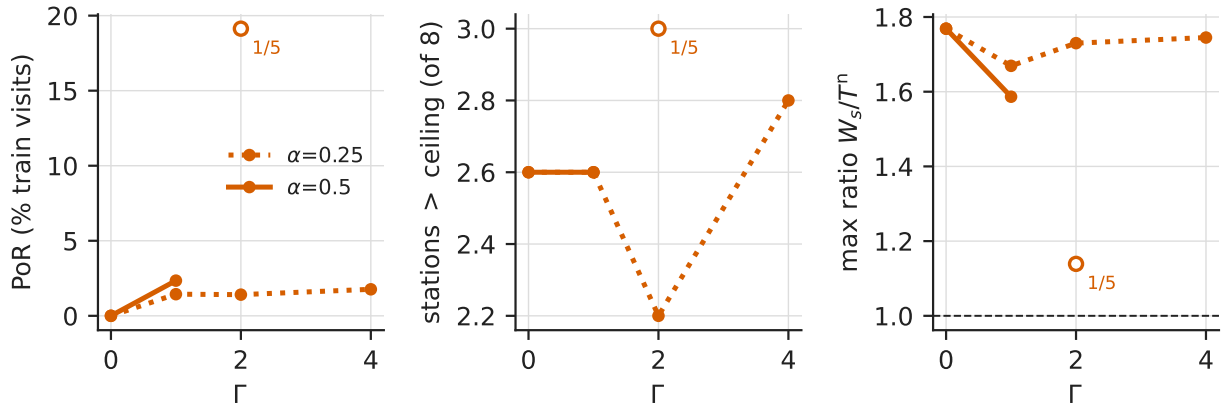


Figure 7: Trade-off along the budget Γ (means over cuts; open markers = arms feasible on fewer than five cuts, annotated with their coverage). Left: price of robustness in train visits. Center: stations above the normalized ceiling. Right: peak load ratio; the dashed line is contract feasibility. Partial insurance moves the peak, not the violation count.

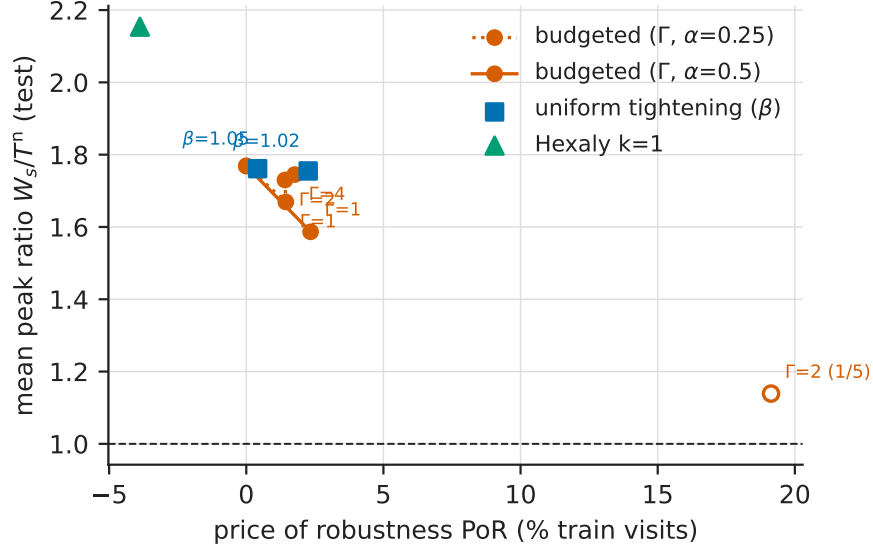


Figure 8: Price-severity frontier: mean out-of-sample peak ratio versus PoR. Budgeted arms (vermillion) move down-right; uniform tightening (blue squares) moves right *without* moving down; the Hexaly reference (green) shows that a stronger visit optimizer concentrates load harder. The open marker is the single-cut $\Gamma=2$, $\alpha=0.5$ arm.

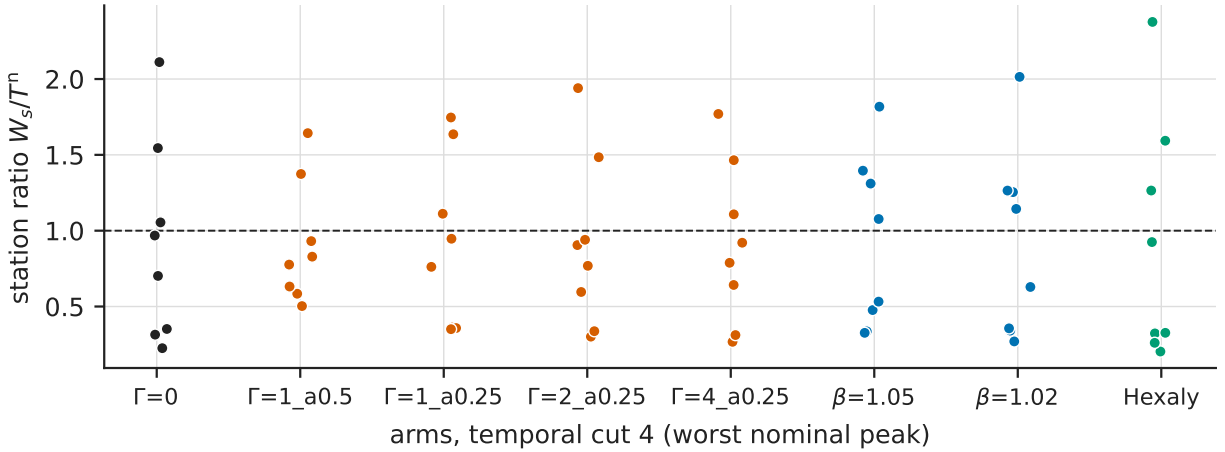


Figure 9: Per-station load ratios W_s/T^n on temporal cut 4 (the worst nominal peak), by arm. The budget compresses the top of the distribution — the station a drifting SKU overloads — rather than shifting all stations uniformly.

5.3 Results

(i) Cheap partial insurance works on severity. The efficient point of the sweep is $\Gamma = 1$ at $\alpha = 0.5$: for a 2.34% visit price it cuts the mean out-of-sample peak from 1.77 to 1.59 and the excess workload fraction X from 17.4% to 13.0% (Table 12). The reduction is *paired sign-consistent*: the peak ratio falls in five cuts out of five ($1.49 \rightarrow 1.47$, $1.48 \rightarrow 1.34$, $1.81 \rightarrow 1.61$, $1.95 \rightarrow 1.87$, $2.11 \rightarrow 1.64$), as does X . Halving the insured magnitude again ($\alpha = 0.25$) roughly halves both the price and the effect.

(ii) Equal price, uniform tightening buys nothing. The $\beta = 1.02$ competitor pays essentially the same price as the efficient budgeted arm (2.26% vs. 2.34%) and leaves the mean peak at 1.75 — indistinguishable from the nominal 1.77 — with per-cut changes that are *sign-inconsistent* (the peak worsens on two of five cuts). The mechanism is visible in Figure 9: tightening rebalances *nominal* loads, but the out-of-sample peak is created by large deviators, and a uniformly tighter ceiling does nothing to separate them — the budget constraint does exactly that, because co-locating two large $\hat{L}_p$ makes the station’s top- Γ protection unaffordable. On this instance family, at equal visit price, **targeted budgeted protection dominates uniform tightening on peak severity.**

(iii) The feasibility wall is sharp. One step further, $\Gamma = 2$ at $\alpha = 0.5$, is constructible on a single cut — and there it nearly restores contract feasibility (peak 1.14, $X = 3.1\%$) at a 19.1% visit price. Insurance beyond the slack headroom is not gradually more expensive; it is discontinuously so, which is the operational face of the arithmetic in Section 5.2.

(iv) Severity, not incidence. Violation *counts* barely move across implementable arms (2.2–2.8 of 8; the count can even rise while the peak falls, as on cut 3). This is exactly what Section 3.8 predicts: M3 underestimates the realized drift, so ex-ante-calibrated protection trims how far stations overshoot, not whether some station somewhere overshoots. Restoring feasibility outright would need the drift-scale magnitudes ($c_1(0.9) \approx 1.2$ –1.6) that no train-internal model provides.

(v) The visit–fragility trade-off, priced. The Hexaly reference beats our incumbents by 3.88% on train visits (and 6.16% on test) — and carries the worst workload profile of the table (peak 2.15, $X = 23.1\%$). Better visit optimization concentrates correlated, drifting SKUs; the budget constraint is the first dial in this project that prices that concentration explicitly instead of discovering it out of sample.

5.4 Solver honesty

Three caveats bound the claims. First, incumbents are metaheuristic: on the truncated support objective of cut 0 the HiGHS dual bound is 30,312 against an incumbent of 43,158 ($\approx 30\%$ gap, largely inherited from the weak LP relaxation of the visit objective), so PoR magnitudes carry solver noise — the paired, sign-consistent within-cut deltas are the reliable signal, not third digits. Second, HiGHS wall-times overran its `time_limit` option on this model (up to 820s against a 480s limit in the solver logs; persisted arm times reach 465s against 180s probes), which is why the sweep runs on the deterministic local-search pipeline and reserves HiGHS for certificates and bounds. Third, $G < 0$ throughout: every robust arm pays its visit price out of sample as well (-1.6% to -4.4% at the implementable budgets). Nothing here claims a visits benefit — the insured quantity is workload-feasibility risk, in line with the falsified-H1 lesson that opened this report.

6 Discussion and recommendations

When is the robust workload constraint worth it? The evidence separates three regimes. (i) *Stationary demand, matched volume* (iscf480 CV): the nominal constraint is adequate — normalized violations exist only because placement sits at the exact slack edge, and they are immaterial (peak ratio ≤ 1.016 , $X \leq 0.40\%$). Robustness would buy little here. (ii) *Temporal drift* (all temporal families, and the planted demand shift): the nominal constraint fails materially and systematically — 1–4 of 8 stations, peak ratios up to 2.52 on iscf480 and 2.63 on the full-catalogue k grid (3.45 in the sparse-train variants), 7–37% of workload above the fair-share ceiling — and the failure is invisible to the as-run check whenever $B < 1/\sigma$. This is the regime the counterpart of Section 4 targets, and the calibration says protection is cheap in budget terms: the realized overloads are covered at median Γ^{cover} of 8/60 and 19/541 products under the best-fitting magnitude model — strikingly close to the $\Gamma \sim \sqrt{n}$ rule of thumb of Remark 2 ($\sqrt{60} \approx 7.7$, $\sqrt{541} \approx 23.3$) — and deviations are head-concentrated (Gini 0.63/0.82), the Bertsimas–Sim sweet spot. (iii) *Volume-artifact regimes* ($B \gg 1$ or $B \ll 1/\sigma$): fix the protocol first; no constraint reformulation can rescue a check that compares incommensurate volumes.

What Γ and what magnitudes the evidence suggests. A per-station budget of order $\sqrt{\zeta_s}$ (reported as the fraction $\Gamma_s/\zeta_s \approx 13\%$ and 4% on the two families) would have covered the median realized overload. Because $\Gamma^{\text{bind}} = 1$ at 58–68% of stations, even $\Gamma = 1$ already reshapes the training solution — the practical risk is over-conservatism, not vacuity, which argues for starting at small budgets and the fractional-budget refinement rather than box protection. The harder open problem is the *magnitude* side: the required $c_1(0.9) \approx 1.2$ – 1.6 (growing with horizon) is far beyond what train-internal variability (M3) predicts on iscf480, so an ex-ante deviation model needs longer demand history or an explicit drift model, not just within-window dispersion; on the full-catalogue instance M3 is a usable starting point (38–80% coverage).

What the implementation shows. The computational study of Section 5 turns the counterpart from a proposal into three practical statements. (a) *Partial ex-ante insurance is cheap and effective on severity*: $\Gamma = 1$ at half the M3 magnitude cuts the out-of-sample peak by ten percent of its value and the excess workload by a quarter, for a 2.3% visit price, sign-consistently across all five cuts. (b) *Uniform tightening is not a substitute*: at the same visit price it leaves the peak where it was, because it rebalances nominal loads without separating the large deviators that create the peak — the budget constraint does that separation by construction. (c) *Full-magnitude protection is a capacity decision, not a modeling one*: the top M3 deviations alone exceed the whole cell’s slack headroom (48.6k vs. 33.0k lines on the first cut), HiGHS certifies infeasibility from $\Gamma = 4$ up, and the one constructible step beyond the headroom ($\Gamma = 2$, $\alpha = 0.5$, one cut) costs 19% of train visits. Insuring the drift this data actually exhibits requires growing the slack σ — a capacity investment that Section 5.2 prices.

Threats to validity. (i) The five temporal cuts share train mass (expanding windows): their spread is a dispersion proxy, not a confidence interval, and per-fold results are correlated. (ii) The volume-normalized ceiling is a *counterfactual protocol choice* (T^{n} re-solves nothing; it asks how the trained layout would fare if the ceiling had been calibrated on the test volume). Other normalizations are defensible; the B diagnosis of the as-run check does not depend on this choice. (iii) Layouts are Hexaly (metaheuristic) placements: balancing exactly to the slack edge is a solver behavior, and better or different placements could shift the borderline CV findings (though hardly

the $1.7\text{--}2.5\times$ temporal overloads). (iv) Instance breadth: all synthetic families derive from one warehouse’s catalogue, and the industrial evidence is a single BERNER cell with a single temporal cut and no persisted layouts (its bindingness verdict rests on assignment bounds). (v) The Γ calibration is retrospective by design; only M3 is available ex ante, and it underestimates drift on iscf480 by an order of magnitude. (vi) The experiment’s incumbents are metaheuristic, with a $\approx 30\%$ HiGHS dual-bound gap on the truncated support objective; paired within-cut deltas are reliable, absolute PoR magnitudes less so. (vii) The partial-insurance grid is coarse ($\alpha \in \{0.5, 0.25\}$), and the infeasible robust arms at $\alpha < 1$ are undetermined-within-budget rather than proven infeasible; only the full-magnitude certificates at $\Gamma \in \{4, 8, 16\}$ are proofs.

A Full per-fold tables

Table 13: iscf480, all stored (split, fold, k) cells. **asrun** = stored **wl_broken** (reproduced exactly); $\text{ratio}^n = \text{peak } W_s^{\text{te}}/T_s^n$; imb. = imbalance. Source: `results/iscf480/per_fold.csv`.

Split	Fold	k	as-run	norm	ratio ⁿ	imb.	B	X (%)
CV	r0f0	1	0	2	1.016	1.117	0.228	0.24
CV	r0f0	2	0	2	1.014	1.116	0.228	0.25
CV	r0f1	1	0	0	0.999	1.098	0.227	0.00
CV	r0f1	2	0	0	0.994	1.094	0.227	0.00
CV	r0f2	1	0	2	1.005	1.105	0.227	0.13
CV	r0f2	2	0	1	1.006	1.106	0.227	0.08
CV	r0f3	1	0	3	1.012	1.114	0.227	0.40
CV	r0f3	2	0	1	1.004	1.105	0.227	0.06
CV	r0f4	1	0	1	1.000	1.100	0.227	0.00
CV	r0f4	2	0	0	0.999	1.099	0.227	0.00
temp.	r0f0	1	4	3	1.714	1.886	1.005	11.02
temp.	r0f0	2	2	2	1.541	1.696	1.005	13.37
temp.	r0f1	1	2	3	1.906	2.097	0.693	19.81
temp.	r0f1	2	1	4	1.710	1.881	0.693	14.75
temp.	r0f2	1	1	3	2.255	2.480	0.457	26.61
temp.	r0f2	2	0	3	1.981	2.179	0.457	21.19
temp.	r0f3	1	0	2	2.519	2.771	0.281	27.23
temp.	r0f3	2	0	3	2.189	2.408	0.281	28.37
temp.	r0f4	1	0	3	2.376	2.614	0.125	30.72
temp.	r0f4	2	0	3	1.882	2.070	0.125	30.25

References

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Table 14: iscf10kt temporal variant, full k grid. Source: `results/iscf10kt/per_fold.csv`.

Cut	k	as-run	norm	ratio ⁿ	imb.	B	X (%)
0	1	2	2	1.50	1.65	0.79	10.6
0	2	2	2	1.47	1.61	0.79	11.8
0	3	1	2	1.56	1.71	0.79	8.6
0	4	2	2	1.32	1.46	0.79	8.3
0	6	1	1	1.49	1.64	0.79	6.8
1	1	1	2	1.69	1.86	0.55	15.6
1	2	0	2	1.53	1.69	0.55	11.8
1	3	0	1	1.56	1.72	0.55	7.7
1	4	1	2	1.65	1.82	0.55	9.9
1	6	0	3	1.44	1.58	0.55	10.5
2	1	0	1	2.15	2.36	0.37	15.8
2	2	0	2	1.66	1.83	0.37	14.2
2	3	0	2	2.06	2.27	0.37	19.2
2	4	0	2	1.90	2.09	0.37	13.0
2	6	0	3	1.75	1.92	0.37	13.0
3	1	0	3	1.93	2.12	0.22	24.6
3	2	0	1	2.36	2.60	0.22	18.8
3	3	0	2	2.16	2.38	0.22	24.5
3	4	0	3	1.78	1.96	0.22	22.6
3	6	0	3	1.68	1.85	0.22	20.0
4	1	0	3	2.35	2.59	0.11	25.4
4	2	0	2	2.63	2.90	0.11	36.8
4	3	0	2	2.26	2.49	0.11	28.1
4	4	0	3	2.55	2.81	0.11	28.2
4	6	0	3	2.03	2.23	0.11	22.2

Table 15: iscf10kt sparse-train variants, $k=1$ rows (full grid, $k \in \{2, 3, 4, 6\}$ included, on disk). Source: `results/iscf10kt/per_fold.csv`.

Variant	Fold	as-run	norm	ratio ⁿ	imb.	B	X (%)
t500	0	8	1	1.78	1.96	8.18	10.8
t500	1	8	1	1.90	2.09	6.61	12.4
t500	2	8	1	2.30	2.53	4.91	17.8
t500	3	8	3	2.14	2.36	3.50	19.8
t500	4	4	2	3.45	3.80	2.07	35.5
t1000	0	8	2	1.56	1.72	4.03	11.1
t1000	1	8	1	1.84	2.02	3.28	11.5
t1000	2	8	2	2.09	2.30	2.53	15.2
t1000	3	6	3	1.71	1.89	1.77	18.4
t1000	4	2	2	3.44	3.79	1.06	35.4
t2000	0	7	2	1.54	1.70	1.97	9.2
t2000	1	7	3	1.49	1.63	1.68	13.2
t2000	2	4	3	1.68	1.85	1.27	15.6
t2000	3	3	3	2.08	2.29	0.90	22.5
t2000	4	1	2	2.69	2.96	0.52	31.0

Table 16: iscfco base arms (unshifted $\rho = 0$ evaluation). $k=1$ metrics are identical across the mm/ms variants by construction. Per-fold $B = 0.858, 0.573, 0.374, 0.221, 0.099$ for cuts 0–4. Source: `results/secondary/per_fold.csv`.

Variant	Cut	k	as-run	norm	ratio ⁿ	imb.	X (%)
mm	0	1	0	3	1.059	1.165	1.46
mm	0	10	0	1	1.028	1.131	0.39
mm	1	1	0	2	1.055	1.161	1.38
mm	1	10	0	2	1.055	1.160	1.20
mm	2	1	0	2	1.045	1.150	1.20
mm	2	10	0	2	1.060	1.167	1.17
mm	3	1	0	2	1.089	1.198	1.27
mm	3	10	0	2	1.064	1.171	1.24
mm	4	1	0	3	1.057	1.164	1.96
mm	4	10	0	3	1.083	1.193	1.52
ms	0	1	0	3	1.059	1.165	1.46
ms	0	10	1	1	1.061	1.167	0.83
ms	1	1	0	2	1.055	1.161	1.38
ms	1	10	0	2	1.064	1.171	1.15
ms	2	1	0	2	1.045	1.150	1.20
ms	2	10	0	1	1.014	1.116	0.20
ms	3	1	0	2	1.089	1.198	1.27
ms	3	10	0	3	1.045	1.149	0.91
ms	4	1	0	3	1.057	1.164	1.96
ms	4	10	0	2	1.089	1.199	1.50

- [4] A. Ben-Tal, L. El Ghaoui, and A. Nemirovski. *Robust Optimization*. Princeton University Press, Princeton, NJ, 2009.

Table 17: Planted-affinity informative arms (reps r0, r2; rep r1 is degenerate as-run with empty layouts and is omitted). The as-run 7–8/8 counts are volume artifact ($B > 1$). Source: `results/secondary/per_fold.csv`.

Arm	k	as-run	norm	ratio ⁿ	imb.	B	X (%)
combine $\alpha=0.0$ r0	1	8	2	1.071	1.180	1.81	1.93
combine $\alpha=0.0$ r0	10	8	1	1.065	1.173	1.81	0.89
combine $\alpha=0.5$ r0	1	8	2	1.049	1.155	3.18	1.05
combine $\alpha=0.5$ r0	10	8	2	1.052	1.158	3.18	0.88
combine $\alpha=1.0$ r0	1	8	2	1.063	1.170	4.40	1.25
combine $\alpha=1.0$ r0	10	8	3	1.055	1.161	4.40	1.83
reassign $\alpha=0.0$ r0	1	8	2	1.092	1.202	1.82	2.42
reassign $\alpha=0.0$ r0	10	8	3	1.093	1.203	1.82	1.89
reassign $\alpha=0.5$ r0	1	8	2	1.170	1.288	1.93	3.43
reassign $\alpha=0.5$ r0	10	8	2	1.180	1.298	1.93	4.51
reassign $\alpha=1.0$ r0	1	8	1	1.165	1.282	1.79	2.27
reassign $\alpha=1.0$ r0	10	8	2	1.196	1.317	1.79	3.68
combine $\alpha=0.0$ r2	1	8	3	1.130	1.243	1.85	2.62
combine $\alpha=0.0$ r2	10	7	2	1.129	1.242	1.85	2.03
combine $\alpha=0.5$ r2	1	8	2	1.039	1.144	3.26	0.79
combine $\alpha=0.5$ r2	10	8	3	1.071	1.179	3.26	1.68
combine $\alpha=1.0$ r2	1	8	2	1.064	1.171	4.48	1.12
combine $\alpha=1.0$ r2	10	8	3	1.035	1.139	4.48	1.13
reassign $\alpha=0.0$ r2	1	8	2	1.128	1.243	1.84	2.04
reassign $\alpha=0.0$ r2	10	8	2	1.124	1.237	1.84	2.00
reassign $\alpha=0.5$ r2	1	8	2	1.196	1.317	1.88	3.84
reassign $\alpha=0.5$ r2	10	7	4	1.244	1.370	1.88	7.40
reassign $\alpha=1.0$ r2	1	8	3	1.150	1.265	1.90	2.78
reassign $\alpha=1.0$ r2	10	8	3	1.158	1.274	1.90	5.54

Table 18: Computational study (Section 5): all feasible (cut, arm) cells on iscf480 temporal. PoR and G are paired against the cut’s $\Gamma=0$ arm; viol. = stations above T^n (of 8); peak = $\max_s W_s^{te}/T^n$. Arms not shown are undetermined-within-budget ($\Gamma=2$, $\alpha=0.5$ on cuts 1–4; $\Gamma=4, 8$ at $\alpha=0.5$ and $\Gamma=8$ at $\alpha=0.25$ on all cuts; $\Gamma=1$ at $\alpha=1$ on cut 0) or proven infeasible ($\alpha=1$, $\Gamma \in \{4, 8, 16\}$, cut 0).

Source: `results/experiment/results.csv`.

Cut	Arm	PoR (%)	G (%)	viol.	peak	X (%)
0	$\Gamma=0$	0.00	0.00	2	1.489	10.6
0	$\Gamma=1$, $\alpha=0.5$	2.11	−3.49	1	1.468	6.4
0	$\Gamma=1$, $\alpha=0.25$	1.70	−3.49	2	1.391	8.6
0	$\Gamma=2$, $\alpha=0.5$	19.13	−23.14	3	1.139	3.1
0	$\Gamma=2$, $\alpha=0.25$	1.21	−2.47	3	1.408	8.7
0	$\Gamma=4$, $\alpha=0.25$	2.02	−1.89	3	1.429	9.4
0	tight $\beta=1.05$	0.64	−0.60	2	1.519	9.2
0	tight $\beta=1.02$	3.10	−3.72	2	1.498	8.3
0	Hexaly $k=1$	−3.45	3.41	3	1.714	11.0
1	$\Gamma=0$	0.00	0.00	3	1.482	14.0
1	$\Gamma=1$, $\alpha=0.5$	2.39	−4.06	3	1.342	10.7
1	$\Gamma=1$, $\alpha=0.25$	1.39	−2.53	3	1.517	10.9
1	$\Gamma=2$, $\alpha=0.25$	1.66	−1.44	2	1.710	11.5
1	$\Gamma=4$, $\alpha=0.25$	0.14	−1.78	3	1.540	9.4
1	tight $\beta=1.05$	0.32	0.25	2	1.533	12.4
1	tight $\beta=1.02$	0.91	−1.87	2	1.427	11.0
1	Hexaly $k=1$	−5.05	7.09	3	1.906	19.8
2	$\Gamma=0$	0.00	0.00	3	1.814	17.4
2	$\Gamma=1$, $\alpha=0.5$	1.63	−3.02	3	1.613	15.4
2	$\Gamma=1$, $\alpha=0.25$	0.47	0.29	3	1.793	17.3
2	$\Gamma=2$, $\alpha=0.25$	1.53	−1.91	2	1.724	15.8
2	$\Gamma=4$, $\alpha=0.25$	1.32	−1.31	3	1.876	15.4
2	tight $\beta=1.05$	0.35	−1.05	3	1.903	15.4
2	tight $\beta=1.02$	2.28	−0.40	3	1.935	17.2
2	Hexaly $k=1$	−4.18	7.22	3	2.255	26.6
3	$\Gamma=0$	0.00	0.00	2	1.948	21.4
3	$\Gamma=1$, $\alpha=0.5$	2.17	−2.69	4	1.867	18.5
3	$\Gamma=1$, $\alpha=0.25$	1.82	−4.35	2	1.899	19.6
3	$\Gamma=2$, $\alpha=0.25$	0.90	−0.19	2	1.868	22.1
3	$\Gamma=4$, $\alpha=0.25$	2.73	−0.21	2	2.110	23.0
3	tight $\beta=1.05$	−0.42	2.69	3	2.034	25.2
3	tight $\beta=1.02$	1.88	1.87	3	1.901	25.2
3	Hexaly $k=1$	−3.30	6.27	2	2.519	27.2
4	$\Gamma=0$	0.00	0.00	3	2.112	23.5
4	$\Gamma=1$, $\alpha=0.5$	3.43	−8.94	2	1.643	14.0
4	$\Gamma=1$, $\alpha=0.25$	1.83	−2.35	3	1.746	20.5
4	$\Gamma=2$, $\alpha=0.25$	1.76	−2.19	2	1.940	19.6
4	$\Gamma=4$, $\alpha=0.25$	2.62	−6.87	3	1.770	18.5
4	tight $\beta=1.05$	1.17	−2.68	4	1.818	22.0
4	tight $\beta=1.02$	3.11	−4.52	4	2.014	23.1
4	Hexaly $k=1$	−3.41	6.81	3	2.376	30.7